

OFAD GOLEM

GETTING STARTED · SHORTCUTS · VERSION 1.1

16-track groovebox, free software (AGPL) by One For All Distribution.

This manual covers the basics; the instruments and effects are described in detail in the **Catalogue**.

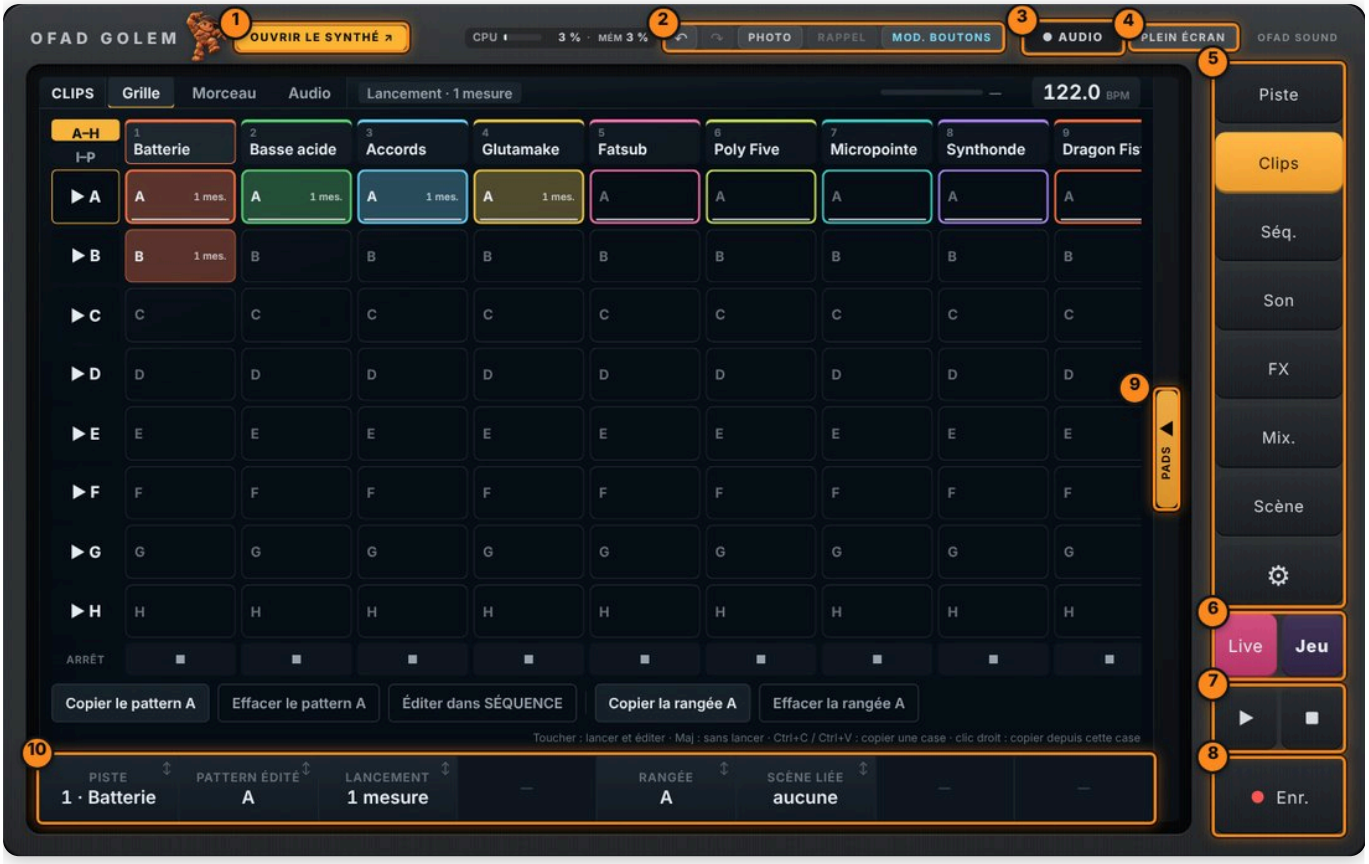
> **The interface is available in English.** Choose the language when you install on Windows, or at the first launch on Android; change it at any time in Settings › Project (**Français / English**). This manual uses the English names of the controls; if you use the French interface, the table below gives the French label you will see on screen.

In this manual	On screen (French)
Screens TRACKS, CLIPS, SEQUENCE, SOUND, EFFECTS, MIXER, SCENES, SETTINGS	PISTES, CLIPS, SÉQUENCE, SON, EFFETS, MIXEUR, SCÈNES, RÉGLAGES
Rail: Track, Clips, Seq., Sound, FX, Mix., Scene, ⚙ · Live · Play view · Rec	Piste, Clips, Séq., Son, FX, Mix., Scène, ⚙ · Live · Jeu · Enr.
CLIPS tabs: Grid, Song, Audio	Grille, Morceau, Audio
Transport: Rec, Play, Stop, Fill, Break, Shift	Enr., Lecture, Arrêt, Fill, Break, Maj
Pad roles: Play, Mute, Solo, Scenes, Steps · Pads (fold-out)	Jouer, Mute, Solo, Scènes, Pas · Pads
Open the synth · Photo · Recall · Knob mods · Live view	Ouvrir le synthé · Photo · Rappel · Mod. boutons · Vue live
Assign to… · Free the knob · Learn MIDI	Assigner à… · Libérer le bouton · Apprendre MIDI
Accent · Slide · Locks · Locks only	Accent · Glissé · Verrous · Verrous seuls
Writing: Acid / Locks · Follow · Nudge · Hide / Show	Écriture : Acide / À verrous · Suivre · Décaler · Masquer / Afficher
Edit · Done · Pad active / stopped · Corners · Size · Height · Rotation	Éditer · Terminer · Pavé actif / arrêté · Coins · Taille · Hauteur · Rotation
Rec performance · Rec motion · Record a take · Arrange mode · Loop	Enr. le jeu · Enr. les mouvements · Enregistrer une prise · Mode montage · Boucle

Keyboard and mouse shortcuts

Everywhere	
Space bar	Play / stop
Ctrl + Z	Undo
Ctrl + Y (or Ctrl + Shift + Z)	Redo
F11	Full screen / back to window

Esc	Close the open window: synth panel, pedalboard, instrument or effect picker, presets, menu, home screen
B held	Break: the Break scene while the key is held (and Fill at the same time)
N held	Fill: FILL steps play, NOT FILL steps go silent
Right-click (long press with a finger) on an LED button, Photo, Recall, M / S, rows A to P, sequence patterns A to P...	Assign to the MIDI controller
Touch the **tempo** (status bar, or the Tempo row of the Play view)	Type the tempo (40 to 240 BPM, to a tenth)
Mouse wheel on the mixer	Scroll the tracks left and right (over a control: adjust it)
Long press (right-click) on a track of the TRACKS screen	Sample editor (tracks that play samples)
Long press on "● Audio"	Start the take right away, with the last settings of the AUDIO screen
Drag a **point of a curve** (SOUND screen)	Adjust the envelope, filter or waveform it shows



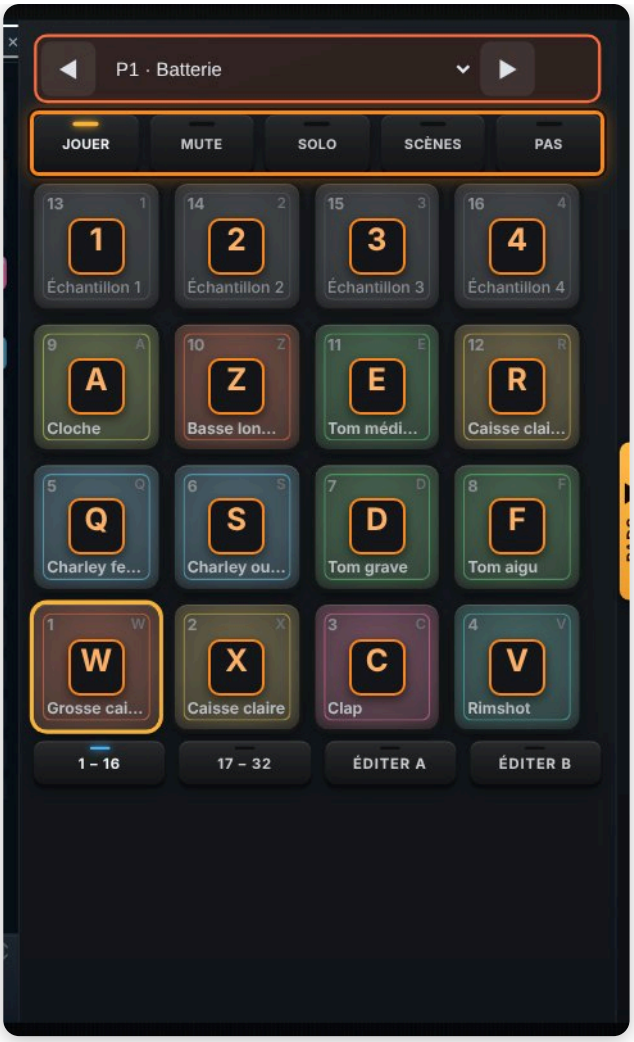
Where the main controls are

1. **Open the synth**: the track's instrument panel, full size.
2. **Undo / Redo** (Ctrl + Z / Ctrl + Y), **Photo** and **Recall** of all settings, **Knob mods**.
3. **Audio**: the AUDIO screen — record the master, tracks, returns or the input, with an effects chain (long press: start right away).
4. **Full screen** (F11).
5. The **rail**: the **8 screens** (Track, Clips, Seq., Sound, FX, Mix., Scene, ⚙ Settings).
6. **Live** (live view: crossfader or vector, 16-track console) and **Play view** ("Jeu": pads, 16 knobs, Fill / Break, crossfader, tempo).
7. **Play** and **Stop** (space bar).
8. **Rec**: record into the pattern.
9. **Pads**: fold out the 16 pads on the right of the screen, on every screen.
10. The **values at the bottom** of the screen: drag ↑↓ to adjust, long press for the menu.

On a phone, undo / redo and **Audio** are in each screen's status bar; Photo, Recall and Knob mods are in the **Play view**.

Pads on the computer keyboard (AZERTY)	
1 2 3 4	13 to 16 (top row)
A Z E R	9 to 12
Q S D F	5 to 8
W X C V	1 to 4 (bottom row)
Shift + key	Accented hit (full velocity)

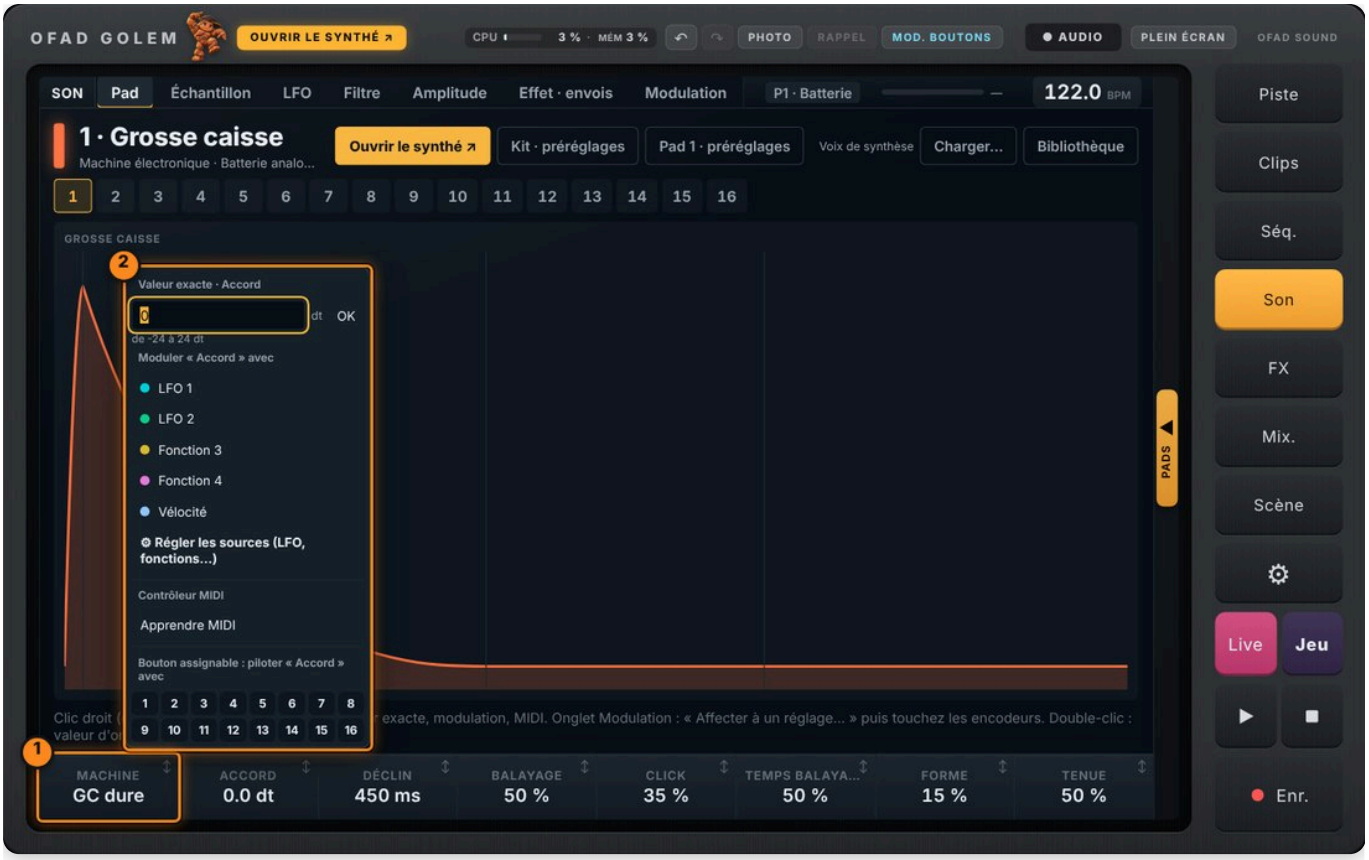
The keys follow the chosen pad role (Play, Mute, Solo, Scenes, Steps), whether the pads are folded out or not.



The keyboard key on each pad (the same small letter is shown in the top right corner of each pad)

Knobs and values (synth panel knobs, values at the bottom of the screens, assignable knobs)	
Drag up / down	Adjust
Shift + drag	Fine adjustment
Mouse wheel	One step (**Shift** + wheel: fine step)
Double-click (double tap with a finger)	Default value
Arrows ↑ ↓ ← → (knob selected)	One step (**Shift**: fine step)
Right-click (long press with a finger)	Menu: exact value, modulation, Learn MIDI, assignable knob (and macros, on a synth panel)
Mixer: double-click on a fader	Default value; double-click on a group name: rename it
Crossfader: double-click	Back to the middle

The values at the bottom of the screens (Swing, Length, Division, X, Y, Crossfader, Scene A...) are adjusted by dragging on them. A **long press** (right-click on PC) opens their menu, the same everywhere: exact value (list of choices or a number to type), **Learn MIDI**, **drive with knob 1 to 16**; modulation only appears for a sound, effect or mix setting. The synth **macros** are only offered on a synth panel.



Adjusting a value and handing it to a knob

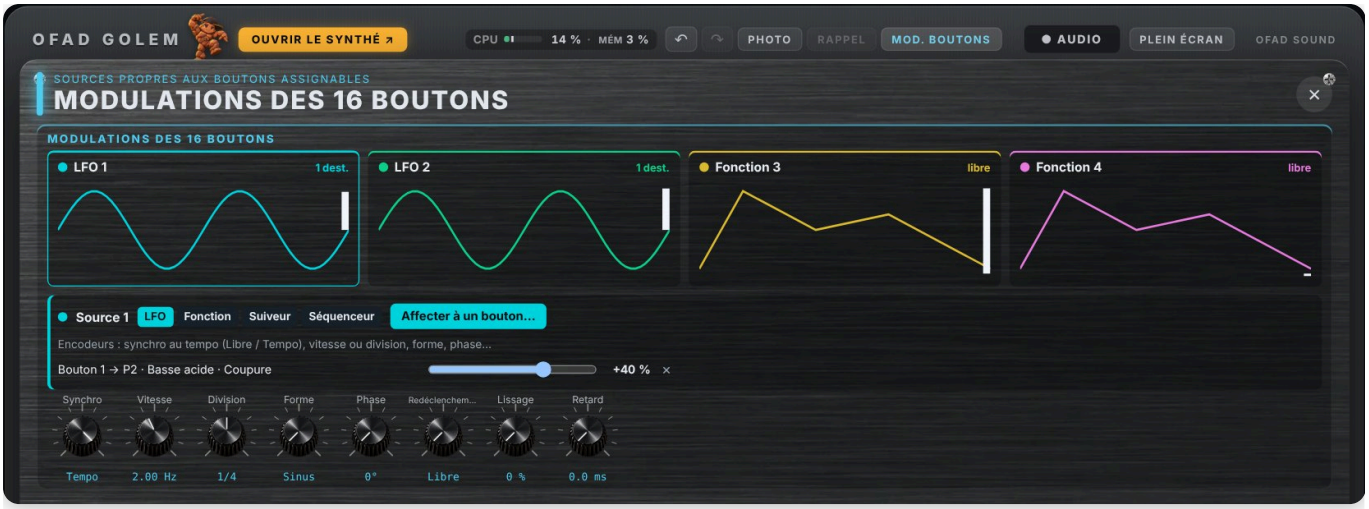
1. A **value at the bottom of the screen**: drag, double-click.
2. Its **menu** (long press, right-click): exact value, modulation, Learn MIDI, knob 1 to 16.
3. The **16 assignable knobs**.

The 16 assignable knobs replace the old encoders: they no longer change with the screen, each one keeps the value you gave it. On PC and tablet they sit in one row of 16; on a phone, in two rows of 8. A **blue diamond** marks them: hollow when the knob is free, filled when it drives something (its name is written under the number). The full travel, from 0 to maximum, takes **a single gesture**.



Long press on an assignable knob

- **Long press** (right-click) on a knob › **Assign to…**: the knob **blinks blue** and everything that can be driven is **underlined in dotted blue**. Touch the value you want: a value at the bottom of a screen, a synth or effect knob, a macro, the crossfader, the tempo, Play, Fill, an M / S… You can change screen or open a synth before touching. **Esc** cancels.
- **Free the knob**: it no longer drives anything.
- **Modulate the knob**: the 16 knobs have **their own modulation sources** (4 sources: LFO, function, follower or sequencer), independent from the synths' ones: a setting can be modulated both by the synth's LFO and by the knob's. The knob menu offers these 4 sources, the **depth** (−100 to +100 %) and "Remove modulation"; a coloured dot shows the source and the knob moves with it. Removing the modulation or freeing the knob puts the setting back as it was. Everything the knob drives can be modulated (sound, effects, mix, crossfader, vector pad, macros), except the tempo and the commands.
- **Knob mods** (next to Photo / Recall, at the top on PC, under the 16 knobs on phone and tablet) opens the panel of these 4 sources, the same as a synth's modulations: shape, tempo sync, speed, phase…; **Assign to a knob…** then touch the knobs to modulate.
- **Learn MIDI**: a knob or fader on the controller turns this knob.



Knob mods: the 4 sources of the 16 knobs



Knob 3 is waiting: touch the value it should drive

Values that belong to one screen (Length, Division, Family…) follow their knob while the interface is open; after opening a project, show the screen concerned once. Sound, effects, tempo, swing, crossfader, scenes, vector pad and macros always respond.

SEQUENCE screen (notes and steps)	
Click on a cell	Add / remove a step (or a note)
P	Brush: dragging writes one step per cell crossed
Ctrl + click	Add to / remove from the selection
Drag from an empty cell	Selection rectangle (**Shift**: adds)

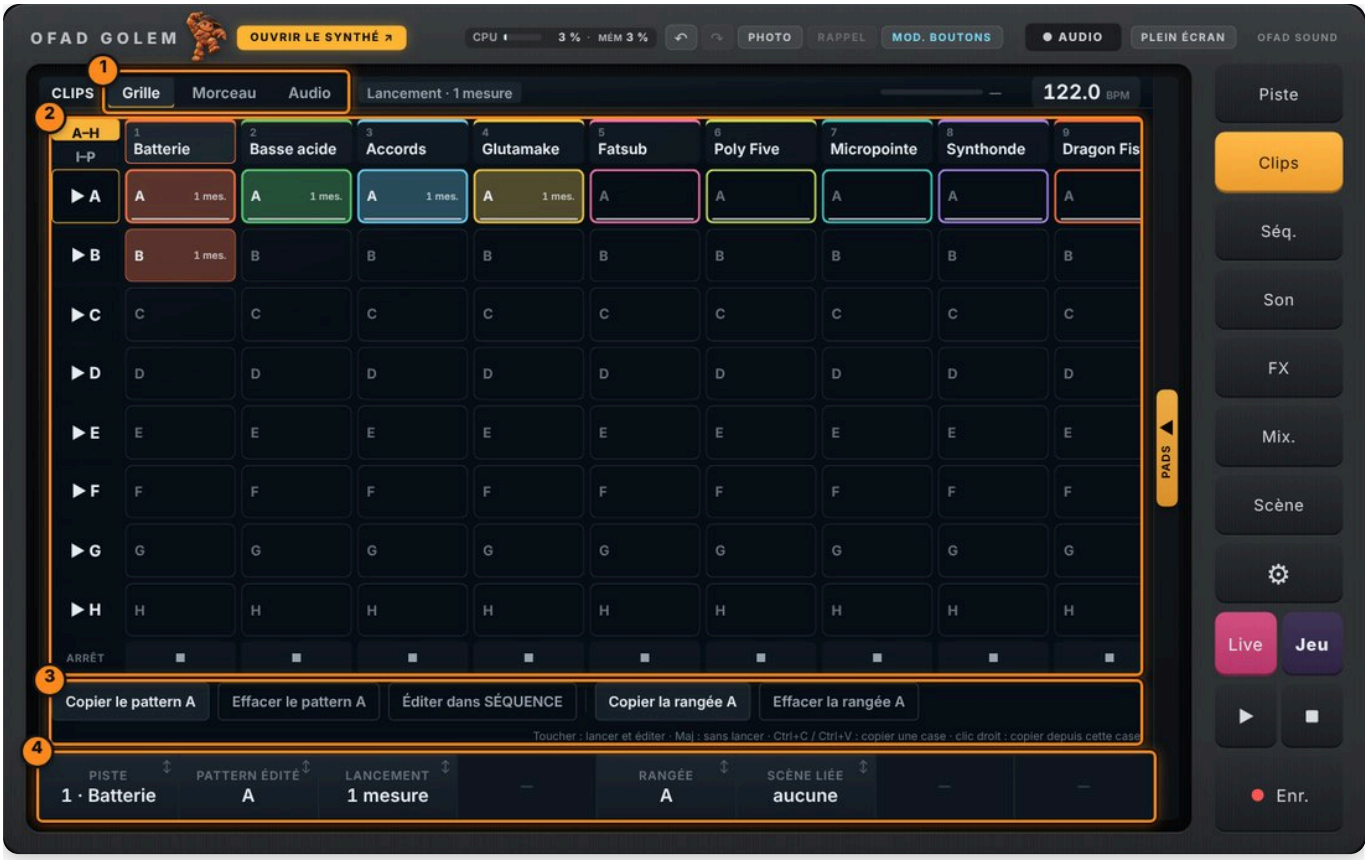
Drag a selection	Move it; **Ctrl** (or Alt) + drag: copy it
Ctrl + A	Select all
Ctrl + C / Ctrl + X / Ctrl + V	Copy / cut / paste under the mouse
Ctrl + D	Duplicate right after
Delete (or Backspace)	Clear the selection
Esc	Empty the selection
Arrows ← →	Move by one step (**Shift**: by one bar)
Arrows ↑ ↓	Note: one semitone (**Shift**: one octave); drums: next pad
Right-click on a step	Step settings (velocity, duration, condition, locks)
Touch an **Accent** / **Slide** cell	Turn the step's accent or slide on / off
Long press (right-click) on an **Accent** / **Slide** cell	The step's value: accent strength, slide time
◀ ▶ next to the pattern (or the small dots)	Turn pages (one page = 16 steps)
Shift + Octave ◀ ▶ (Play view)	Previous / next page
Rec then play	The notes you play are written into the pattern (quantise: Settings)
Touch a **lane name** (drum kit, T1 to T8, T1 to T4)	Select the lane and **hear its sound** while the finger stays down
Long press (right-click) on **A … P** under the sequence	Assign that pattern to the MIDI controller (it launches it on the chosen track)
◀ **Pads**	Fold out the pads next to the sequence to play while writing



The SEQUENCE screen of a melodic instrument (here the acid bass)

1. The **track** shown (◀ ▶ or the list).
2. The **pattern**: number of bars (1 to 32), division (1/16, triplets…), speed.
3. **Turn pages**: ◀ ▶ and "page / total" (dots up to 4 pages).
4. The **Accent** and **Slide** lines (and "Locks only": the step sets its settings without playing). Long press on a cell: its value (see "A step's value").
5. The **piano roll**: each step gets whatever note you want, up to 4 notes per step (chords).
6. **Select**, **Brush**, select all, patterns **A to P** (8 at a time; the small "I–P" / "A–H" chip shows the other page), duplicate, clear.
7. **Notes / Curves**: the knob movements recorded with **REC** in this pattern.
8. The **track settings** when no step is selected (swing, length, speed…) — see "3. Writing a sequence".

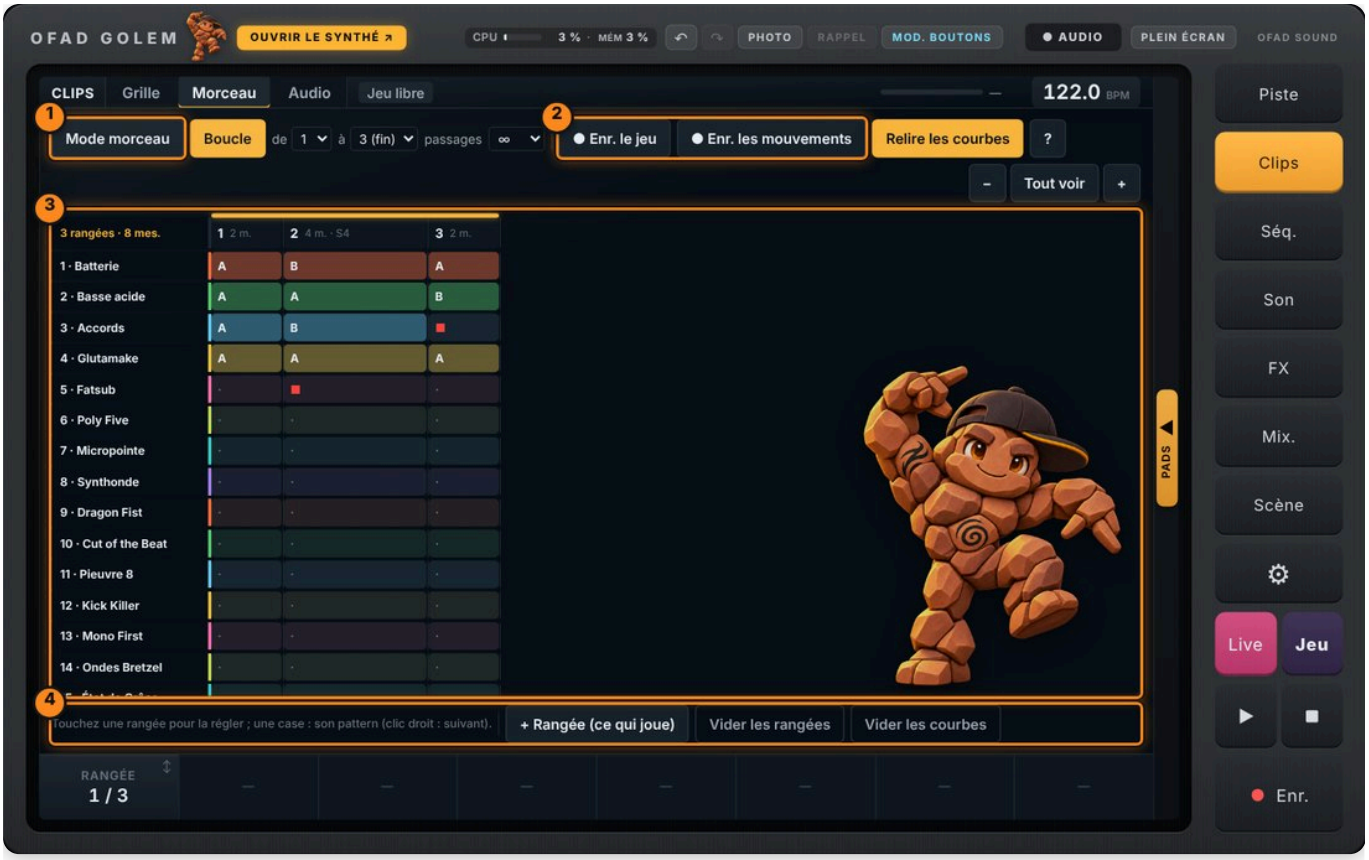
CLIPS screen › Grid	
Touch a cell	Launch and edit the pattern
Shift (button) + touch	Edit without launching
Ctrl + C then a cell, **Ctrl + V**	Copy a pattern to another cell (same track or another track)
▶ at the head of a row	Launch the same letter on all tracks (and the scene linked to the row)



CLIPS › Grid

1. The **Grid**, **Song**, **Audio** tabs.
2. The **grid**: one column per track, one cell per pattern; ▶ launches a row, ■ stops a track. The 16 rows (A to P) are shown 8 at a time: **A–H** / **I–P** in the top left corner.
3. Copy, clear or edit the selected pattern or row.
4. The values at the bottom: track, edited pattern, **launch quantise**, row, **linked scene**.

CLIPS screen › Song and curves	
Ctrl + wheel	Zoom the timeline
Curve editor: **Shift** + drag	Draw a straight line
Curve editor: right-click + drag	Erase

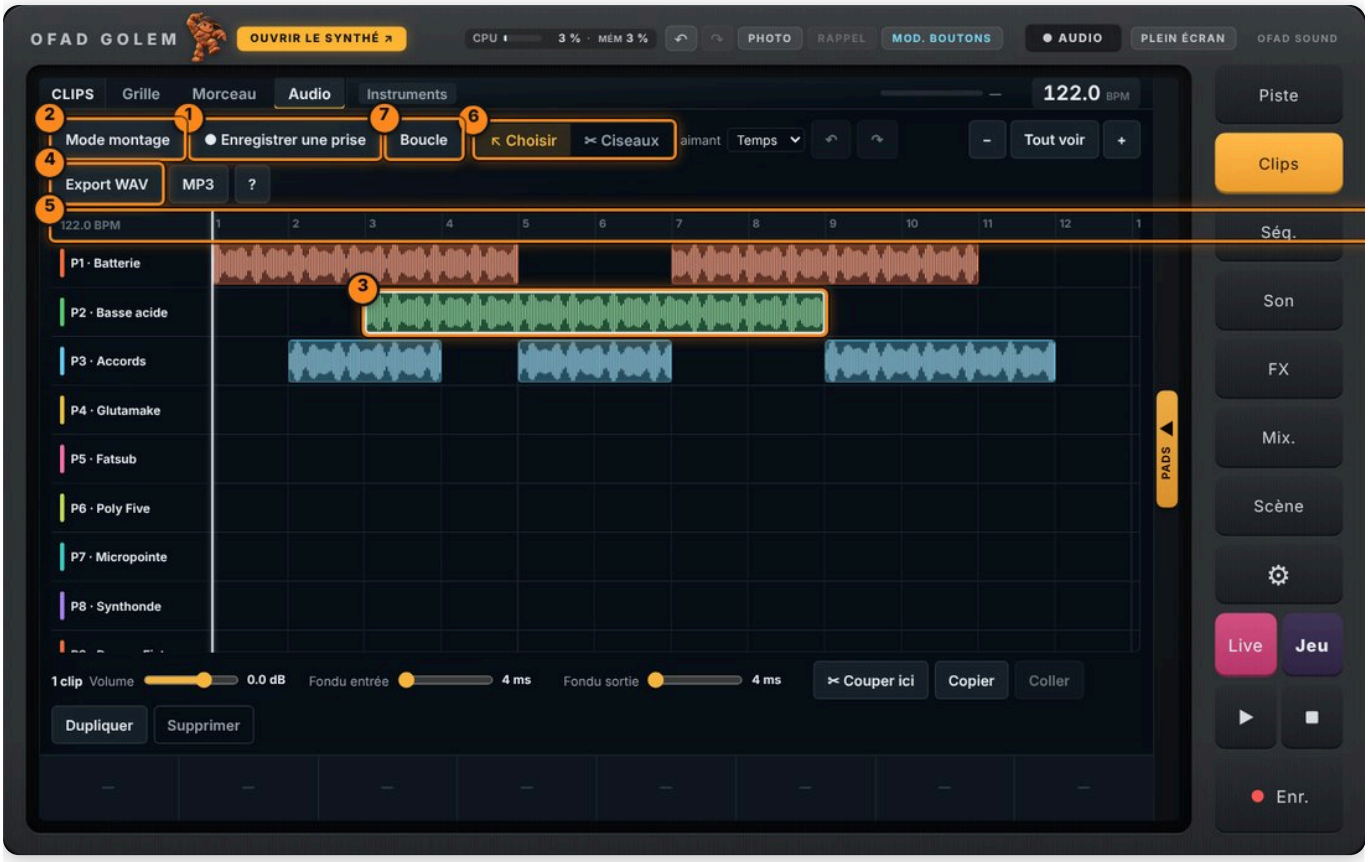


CLIPS › Song

1. **Song mode** (the rows play one after another) or **Loop** (the grid plays).
2. ● **Rec performance** (your playing becomes rows) and ● **Rec motion** (knobs, faders, mutes, solos, crossfader as curves).
3. The **rows** played one after the other; the **curves** below.
4. Add what is playing as a row, clear the rows or the curves.

CLIPS screen › Audio (arrangement)	
Drag a clip	Move it (also to another track)
Drag the edge of a clip	Trim it
Ctrl (or Shift) + click	Select several clips
Drag a rectangle in empty space	Select the clips it touches (**Ctrl**: adds)
Ctrl + A	Select all clips
Right-click on a clip	Menu: split here, copy, duplicate, volume to 0 dB, delete
Right-click in empty space	Menu: paste here, playhead here, select all
✂ tool, or S key	Split (S: at the playhead)
Ctrl + C / Ctrl + V	Copy / paste at the playhead
Ctrl + D	Duplicate right after
Delete (or Backspace)	Delete
Ctrl + Z / Ctrl + Y	Undo / redo (arrangement)
Click on the ruler (drag: move it)	Place the playhead — the next take lands there

Right-click (or **Shift**) + drag on the ruler	New loop
Drag the orange loop bar	Move the loop; its edges: lengthen or shorten it
Loop on + "● Record a take"	The take lasts exactly the loop and **stops by itself** at its end



CLIPS › Audio (arrangement)

1. ● **Record a take** (it lands at the playhead, or at the start of the loop when Loop is on).
2. **Arrange mode**: playback plays the clips instead of the instruments.
3. A **clip**: drag, trim by the edges, right-click: menu.
4. **WAV / MP3 export** of the finished arrangement.
5. The **ruler**: click = playhead; right-click (or Shift) + drag = loop (orange bar).
6. **Select** / **Scissors** tools and snap (bar, beat, 1/16, free).
7. **Loop**: when it is on during a take, it sets the take's length to the step.

Getting started

- At launch, the mascot shows while loading, then the **home screen**: resume the last session, new empty project, **Techno demo** or **Tribe demo** (1.1), open a file, or a project from the folder. (Settings › "Home screen at startup" to show it or not.)
- **Language**: French or English, chosen when installing (Windows) or at the first launch (Android); change it at any time in Settings › Project (**Français / English**).
- Sound card: **Options** (top left of the window) › Audio/MIDI Settings. For the lowest latency: the sound card's ASIO driver or "Windows Audio (Low Latency Mode)", buffer of 128 to 256 samples.

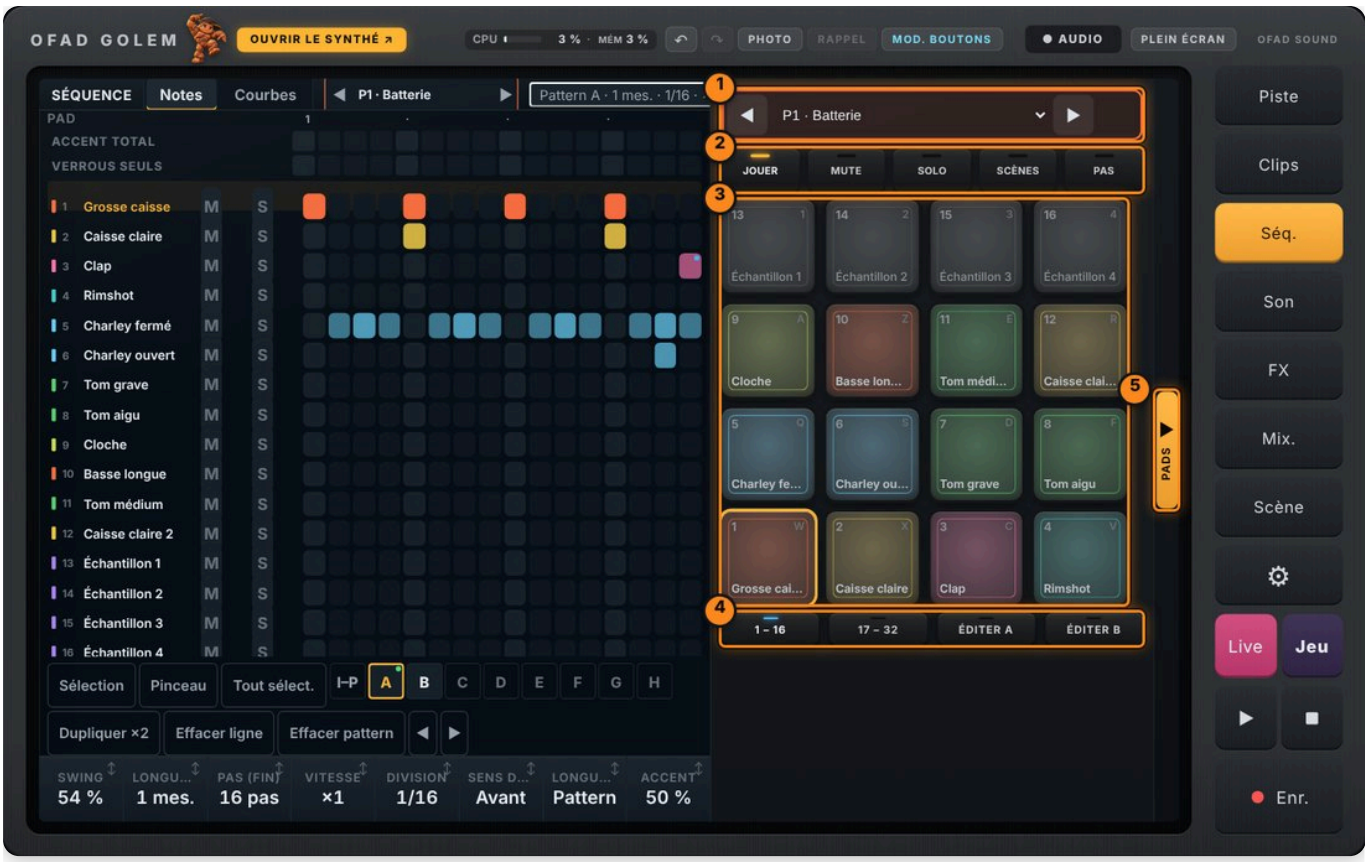


The home screen at launch: resume, new project, demo, open, projects in the folder

The device: a single view

Since 1.0, GOLEM has **a single view**, the same on computer, tablet and phone: **the screen takes all the space** and a **rail** on the right gives everything else.

- **Rail**: the **8 screens** (Track, Clips, Seq., Sound, FX, Mix., Scene, ⚙ Settings), **Live** and **Play view**, **Play / Stop**, and **Rec** at the bottom.
- **Fold-out pads**: the yellow ◀ **Pads** tab, on its own strip at the right edge of the screen, folds out the 16 pads next to the chosen screen, on every screen. Edit the sequence, the mixer or the sound on the left and play the track on the right. ▶ **Pads** folds them back; GOLEM remembers the choice.
- The pad panel: the **played track** (◀ ▶ or list), the **pad role** (Play, Mute, Solo, Scenes, Steps), the 16 pads, the **scene bank** (1 – 16, 17 – 32) and **Edit A / B**. On a drum kit in Mute or Solo, "Drum sounds / 16 tracks" chooses what the pads mute.
- **Values at the bottom** of each screen: adjust them with a finger or the mouse (drag ↑↓), long press for the menu.
- **Top bar** (computer): "Open the synth", CPU load, undo / redo, Photo, Recall, Knob mods, Audio recording, full screen. On a phone, undo / redo and Audio are in each screen's status bar.
- The **computer keyboard** always plays the pads (even folded): space, B, N and the pad keys.

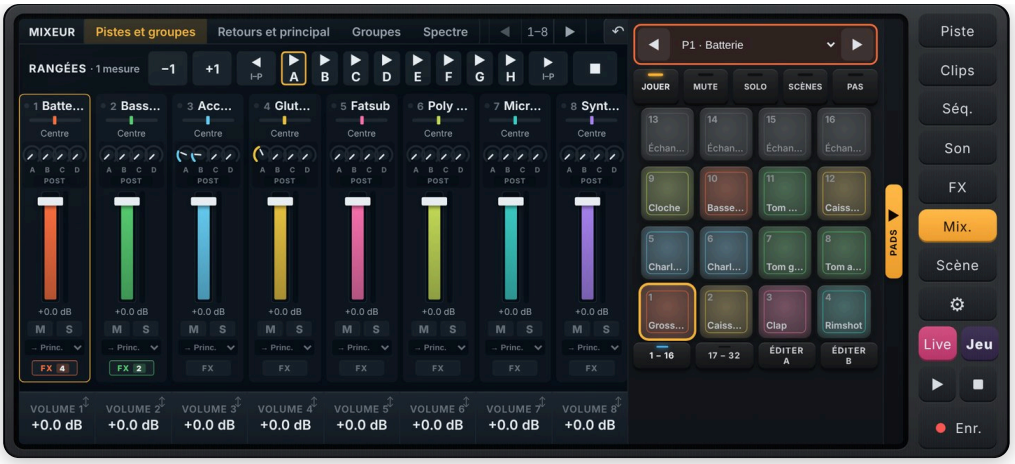


The pads folded out next to the Sequence screen

1. The played track. 2. The pad role. 3. The 16 pads. 4. Scene bank, Edit A / B. 5. Fold the pads back.



Sequence and pads side by side (computer)



The mixer with the pads folded out (phone)

The Play view

Play view ("Jeu" on the rail) replaces the screen with a full playing surface: the played track (◀ ▶), **rows A to H**, the 16 pads and their role, the **16 assignable knobs**, **Photo / Recall / Knob mods**, **Fill, Break, Break 1 bar, Shift**, octave and **crossfader**, the **tempo** (– / value / + / Tap). Touch **Play view** again, or a screen on the rail, to close it.

In the Play view, **Mute** and **Solo** act on the **16 mixer tracks**; each pad keeps its **track colour** (greyed: muted; blue frame: solo). Elsewhere, on a drum kit, the FM Player or Pieuvre 8, they can mute the sounds or the inner tracks.



The Play view

1. Played track. 2. Rows A to H. 3. Pad role. 4. The 16 pads (Mute / Solo: the 16 tracks). 5. The 16 assignable knobs. 6. Photo, Recall, Knob mods. 7. Fill, Break, Break 1 bar, Shift. 8. Octave and crossfader. 9. Tempo.

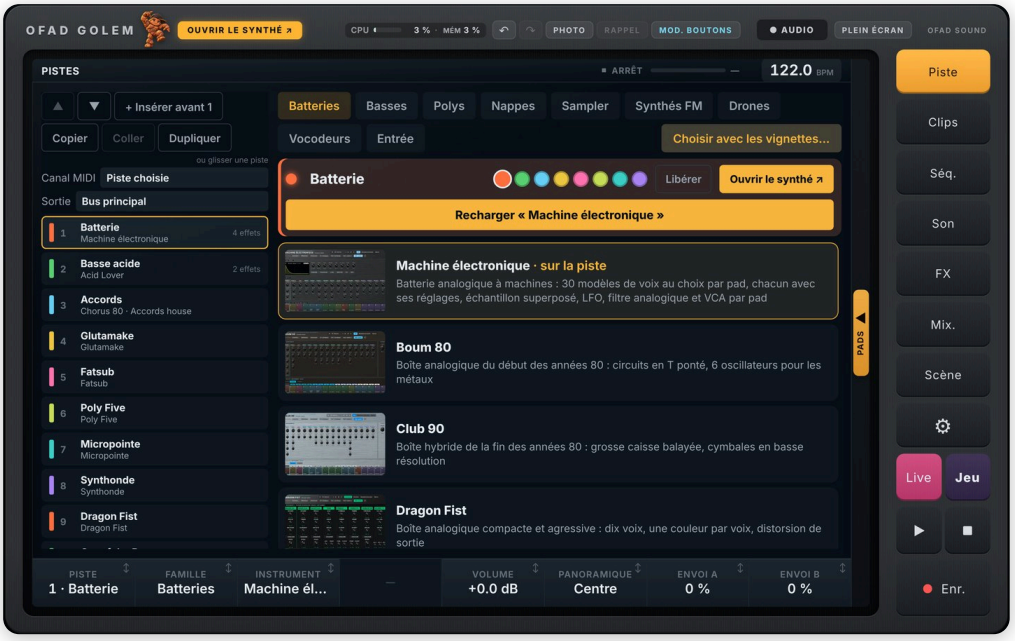


The Play view on a phone

Editing basics

1. Choosing a sound

TRACKS screen: choose the instrument of each track (or "⇄ Instrument" on the panel, with pictures). **Presets** are chosen by category.



TRACKS screen: the instrument, name and colour of each track

2. Playing

The pads, the computer keyboard, the panel's keyboard or a MIDI controller (notes go to the track shown). "Open the synth" shows the instrument panel full size.



An instrument panel (here the drum machine kit): all its settings, its pads or its keyboard

The header of every panel always has three lines, on PC as on a phone: 1. the instrument name, the transport (● ▶ ■), the track selector (◀ ▶, M, S) and ✕ to close; 2. the pages (Sound, Steps, Track modulations, Macros…), which scroll sideways when there are many; 3. the tools: sample and library (if the instrument has them), ⇄ Instrument, FX · Effects, Presets.

3. Writing a sequence

Each track has **16 patterns** (A to P), from 1 to 32 bars. A sequence can be written in three ways, as you like — it is **always the same sequence**: a step set on the panel shows up at once in the SEQUENCE screen, and the other way round.

- **With the mouse**, SEQUENCE screen: one line per pad (drums) or a piano roll (melodic instrument), where **each step gets whatever note you want** (up to 4 notes per step for chords).
- **By playing**: **Rec**, then play the pads, the panel's keyboard, the computer keyboard or the MIDI controller, in time. The notes are written into the pattern while it loops (quantise from 0 to 100 % in Settings; the rest of the timing offset is kept). **REC** also records the **movements** of the track's knobs as curves in the pattern (Curves tab).
- **On the synth panel**, in "Sequence steps", with two writing modes to choose from ("Writing"): **Acid** or **Locks**. The choice is kept, and the same everywhere. On melodic synths, the chosen row is also **above the keyboard**, on every page of the panel: you write your steps without leaving the Sound page. **Hide** ▾ folds it away (more room for the settings), **Show** ▴ brings it back.
- **Step** pad role: the 16 pads set or remove the steps of the page shown, like a drum machine's step programming.



The step row above the keyboard: Writing Acid or Locks, Nudge, Hide

The acid sequencer (basses and synths): the 80s step row, step by step.



Bass panel: "Acid" writing

1. **Writing**: Acid or Locks.
2. The **pages** of 16 steps (one button per page when the pattern is longer).
3. The step's **note**: touch to set or remove the step; drag up or down (or the wheel) to change the note by a semitone.
4. ▲ ▼: one octave up or down.
5. Step **velocity** and **duration** (drag on the value).
6. **Acc.** (accent: the step plays louder and with more "bite", set by the Accent knob) and **Sl.** (slide: the note holds until the next step and glides to it; same note: tied, without retriggering). Touch: on / off; **long press**: the step's value (see below).
7. **FX · Effects**: the track's effects as pedals (see 5).
8. **Sound · Track modulations · Macros**: the other pages of the panel.
9. ⇄ **Instrument**: change instrument with pictures.

A step's menu: a **long press** on the note of a set step opens its menu under the row, in large buttons (handy on a phone): **Accent** and **Slide** (touch: on / off; "...": their amount), **Octave** ▼ ▲, **note**, velocity, length, offset, condition, probability, repeats. The selected step gets an **orange frame** and "**Step 3 selected ×**" lights up above the row: its settings change that step only. The keyboard does not change its note (use **Note** ◀ ▶, or drag the note up or down); it picks the note of the steps you place next. ×, or touching another step, closes the menu.

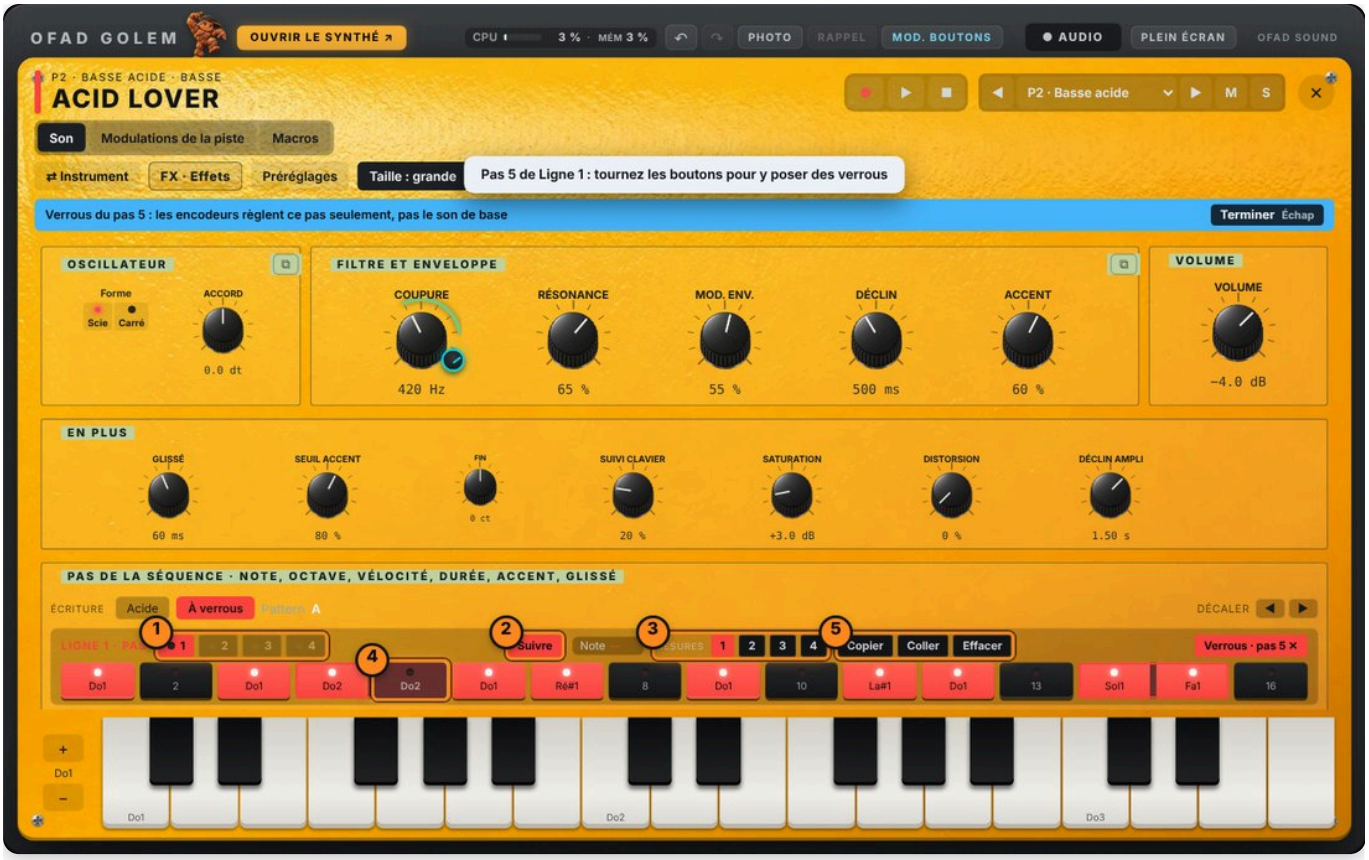


Acid writing: step 5 selected, orange frame, "Step 5 selected ×" lit



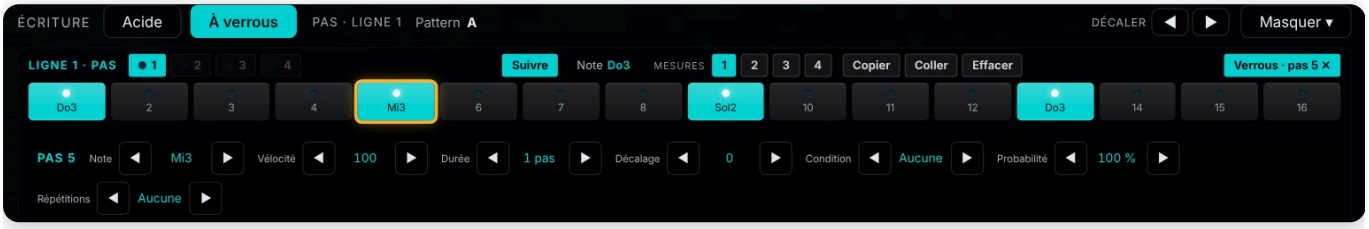
Long press on a step: its menu, with large Accent and Slide

The locks sequencer: each step can keep its own settings (cutoff, resonance, effects...), on top of its note.



Bass panel: "Locks" writing, step 5 selected

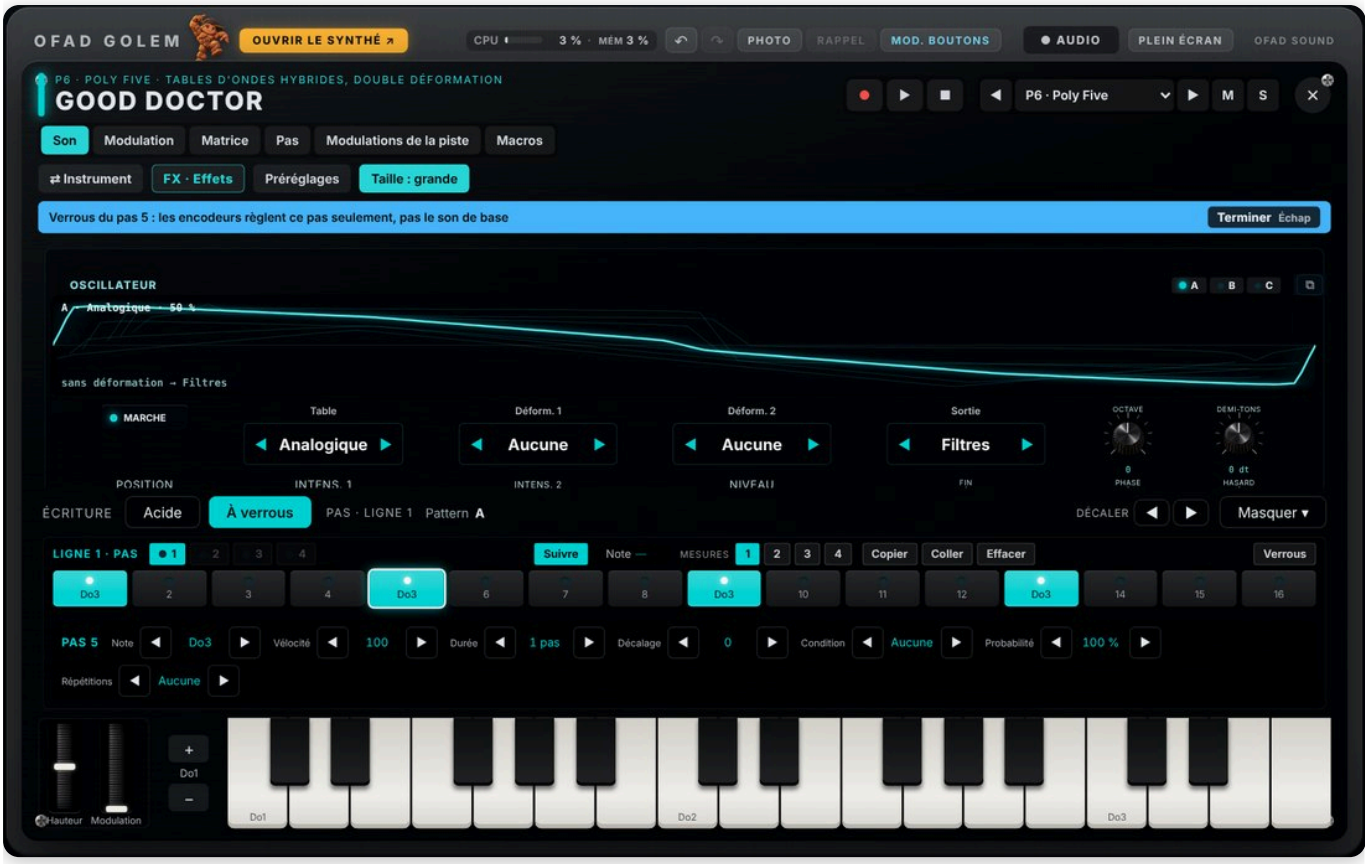
1. **Pages 1 to 4** (16 steps each): touch a page to show it.
2. **Follow**: the page turns by itself with playback (touching a page stops following).
3. **Bars**: length of the line, 1 to 4 bars (the pattern grows if needed).
4. A **selected step** (long press on the step, or the **Locks** button then touch the step): turn any knob on the panel, the pedalboard or the SOUND screen — the value is **locked on that step** only. The step gets an orange frame and the **Locks** button lights up with its number ("**Locks · step 5 ×**", visible even on a phone); a banner at the top reminds you too. Touching Locks, Esc or "Done" releases the step; so does placing or removing a note on another step. In **Acid** writing, the same "Locks · step 5 ×" button lights up above the row while that step is selected.
5. **Copy / Paste / Clear** the locks of the selected step ("No locks": all those of the line).
6. **Glide** and **Mute** (as soon as the line has locks): **Glide** makes the settings travel smoothly from one lock to the next, over the step's length, instead of jumping at the start of the step; **Mute** silences the line's locks without deleting them (the steps play the base sound) — touch **Muted** to bring them back. Both are saved with the project. A lock only plays on a step that holds a note.



"Locks" writing: step 5 selected, "Locks · step 5 ×" button lit

On synths, the locks row writes **the last note played** (pads, the panel's keyboard, or a MIDI keyboard while stopped): it is shown ("**Note** F1") and each step set shows its note instead of its number. A selected step (long press) opens its **menu** under the row: note, velocity, length, offset, condition,

probability, repeats. The keyboard does not change the note of a step already set: **Note** ◀ ▶ in its menu.

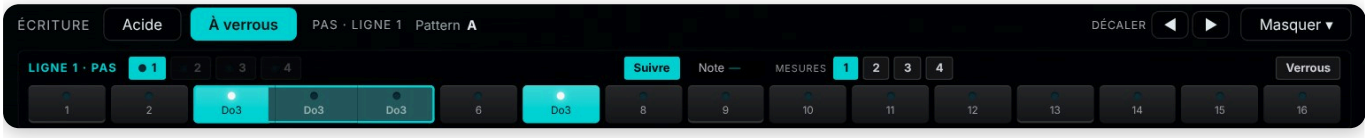


"Locks" writing: the selected step's menu

Note length: in Acid and Locks writing, a note lasting several steps gets a large frame along its whole length, filled inside, and the steps it covers show its note in small. You can see a note is still sounding, and place another note on top. To lengthen or shorten a note: **length** in the step's menu.



Acid writing: a three-step note, framed along its whole length



"Locks" writing: the same frame

Removing a note resets its step: length, offset, accent, slide, condition, probability, repeats and locks. The next note placed there starts from fresh settings.

Nudge ◀ ▶: the whole writing slides one step left or right, wrapping around (the step that falls off comes back on the other side); notes, accents, slides, conditions and locks follow. On the panel: to the right of "Writing" (drums: **All** or the selected **Pad** only), or in the locks row (that line only); in the SEQUENCE screen: ◀ ▶ after "Clear pattern" (**Shift**: the selected line only).

Panel transport: on the first line of every panel, just left of the track selector, ● record, ▶ play (❚❚ to stop), ■ stop — without leaving the instrument.

A step's value: accent and slide. In the SEQUENCE screen (**Accent** and **Slide** lines) as on the panel (**Acc.** and **Sl.**), a touch turns the step's function on or off; a **long press** (right-click on PC) opens its value, full size:

- **Accent**: the strength of this step's accent, from 1 to 100 %. With no value, the step takes the track's accent (the Accent setting at the bottom of the screen).
- **Slide**: the glide time to the next note, from 1 to 100 % of the step (the higher, the longer the glide), on synths that have portamento. With no value: the instrument's portamento.
- ▲ ▼ (held, the value scrolls), **drag** your finger on the value, or **Type...**; **Default** goes back to the track's or instrument's value. Every change is heard at once, and the window stays open while you adjust. Setting a value turns the step's function on.
- A cell that has its own value shows it (for example "54"). Values follow copy, nudge and undo, and are saved in the project.



The accent strength of a step

In the SEQUENCE screen, right-clicking a step opens the same settings: velocity, duration, offset, **note**, condition (probability, A:B, Fill...), repeats, flam, and the **Locks** button.

Turning pages: a page shows 16 steps (one bar in 1/16). In the SEQUENCE screen, ◀ ▶ at the top (or the dots), or **Shift** + Octave ◀ ▶ on the transport; on the panel, the page buttons (and **Follow** in Locks writing); for the pads in Step role, the page shown is the one being written.

Track settings (no step selected)

The values at the bottom of the screen set the track's pattern:

Value	Action
Swing	Shifts every other step (50 %: straight, up to 75 %)

Value	Action
Length	Number of bars of the pattern (1 to 32)
Steps (end)	Length to the step (e.g. 15 steps for an odd loop)
Speed	×1/8 to ×2: the pattern plays slower or faster than the tempo
Division	Length of a step: 1/4, 1/8, 1/16, 1/32, 1/64, or triplets
Direction (drums: **Pad direction**)	Forward, backward, ping-pong, random
Pad length (drums)	A line that loops on its own length (polyrhythms)
Accent	Strength of the steps on the Accent line
Visible octaves (melodic)	The octaves shown in the piano roll

Step settings

A **first click** on a set step **selects** it (it gets a frame); a second click removes it. With the step selected, the values at the bottom of the screen set **that step only**:



A kick drum step selected: its settings at the bottom

1. The **line** (a pad of the kit): touching its name selects it and all its steps (once more: deselects them).
2. **M / S** of the line: mute or solo a line.
3. The **selected step**.
4. The **Accent** lines (here "Total accent": every sound of the step louder) and **Locks only**.
5. The **tools**: Select, Brush, Select all, patterns A to P (right-click: launch it), Duplicate ×2, Clear line, Clear pattern, ◀ ▶ Nudge; for the selected step: "Lock settings on this step", "Close the step", "Remove condition and locks".
6. The **step settings**:

Value	Action
Note (melodic) or **Flam** (drums)	The step's note; or a small hit just before the main one (8 to 60 ms)
Velocity	The **volume** of the hit (1 to 127)
Duration	How long the note holds (from 1/8 of a step to several bars)
Offset	Push the step slightly early or late (± 32/64 of a step): "human" feel
Condition	When the step plays (see the table below)
Probability	Chance that the step plays (0 to 100 %), on top of the condition
Repeats	2 to 8 hits within the step (rolls), at equal, decreasing ↘ or increasing ↗ strength
Locks	Turn it right: the knobs of every screen write into this step

Step sound (1.1): in the **Locks** row under the keyboard, a step's details have a **Sound** list. Pick a preset of the instrument (numbered as for program change: "Init" = 1, the factory sounds bank by bank, then yours): **this step plays with that sound**, and the track goes back to its own sound at the next step without a locked sound. The step shows "S12". Locks on the same step override the chosen sound. "— track's sound —" removes the step sound. On a **MIDI track**, it is the device's sound (program), and the bank is locked too, with the **Bank** list of the top bar (see § 14).



Step sound: step 3 plays the instrument's sound no. 3 ("S3" mark)

Conditions: steps that don't play every time

A condition makes a step play only in certain cases: that is what brings a short loop to life, with variations and small fills without writing other patterns.

Condition	The step plays…
None	every time
A:B (1:2, 2:2, 1:4, 3:4…)	on pass A out of B: 1:2 = every other loop (the 1st), 4:4 = on the 4th loop only
FILL	only while **Fill** is held (or Break): the fills

Condition	The step plays…
NOT FILL	only while Fill is **not** held (what goes silent during the fill)
PRE / **NOT PRE**	if the last conditional step on the same line played / did not play
NEIGHBOUR / **NOT NEIGHBOUR**	if the last conditional step on the line above played / did not play
1ST / **NOT 1ST**	only on the pattern's first loop / on every loop except the first

Fill and Break: live fills and drops

- **Fill** (Play view, live view, or **N** key held): while it is held, the **FILL** steps play and the **NOT FILL** steps go silent. In the same pattern you can prepare a snare fill, a tom roll (with **Repeats** ↗), an open hi-hat… which only appear when you press: mini-breaks on demand.
- **Break** (button, or **B** key held): applies the **Break scene** while it is held — it's a scene like any other (set in the SCENES screen): for example kick muted, filter closed, echo pushed. Break **also triggers Fill**: FILL steps play at the same time.
- **Break · N bars** (timed Break): Break for 1, 2, 4, 8 or 16 beats, then it stops by itself on a beat (**Shift** + press: change the length).
- With **Shift** on, Fill and Break latch instead of being held.

Selecting, clearing, deleting everything



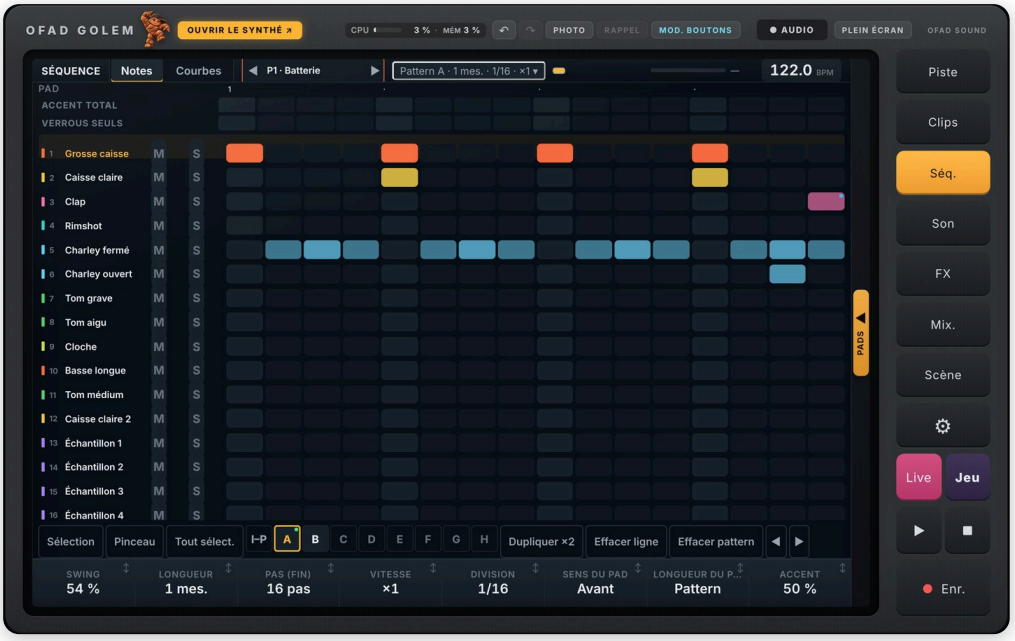
"Select all": the 20 steps of the pattern selected

1. The **selection**: selected steps are framed.
2. The values at the bottom set **all the selected steps at once**: velocity (average; the differences between steps are kept), duration, offset, condition, probability, repeats, locks; on melodic instruments, "Transpose".
3. The **selection bar**: ◀ ▶ (nudge one step), Copy, Cut, Paste, Duplicate, Clear, Copy on drag, Locks, All.

A single selected step (Select mode or rectangle) opens **its menu**, like a chosen step: note, velocity, length, offset, condition, probability, repeats, locks.

The gestures:

- **Select**: **Ctrl + A** or "Select all" (the whole pattern); the name of a line (all its steps); **Ctrl** + click (one step more or less); drag a rectangle from an empty cell; with a finger, the **Select** button then touch the steps.
- **Clear the selection**: **Delete** (or "Clear" in the bar).
- **Clear a line**: "Clear line" (the selected line: the pad, on drums).
- **Delete everything**: "**Clear pattern**" empties the whole pattern shown (notes, conditions, locks); the other patterns are untouched.
- A step keeps its note but loses its conditions and locks: "Remove condition and locks".
- A mistake? **Ctrl + Z** undoes (50 steps).



SEQUENCE screen: one line per pad (drums) or a piano roll (melodic instrument)

4. Launching

CLIPS › Grid screen: each cell launches a pattern (at the start of the next bar); ▶ at the head of a row launches the same letter on all tracks; ■ stops a track.

16 rows, on two pages. Each track has 16 patterns: the rows go from **A to P**. The **Rows** bar (Mixer, Scenes, live view) shows them on a single line, 8 at a time:

- **◀ / ▶** (labelled "I–P" or "A–H") turn the page. They get a yellow outline when the row that is playing is on the other page.
- **−1 / +1** launch the previous or next row and **turn the page by themselves** (from H, +1 goes to I). They **skip empty rows**: if I to P are not written, +1 from H goes back to A.
- The page follows the row launched, also from the MIDI controller or the song, and it is the same everywhere (rows bar, CLIPS grid).
- From a MIDI controller, each of the 16 rows can be learned (right-click on its button), like −1 / +1 and ■.

5. Shaping the sound

One instrument, three views, the same settings. A track's sound is set:

- on the **synth panel** ("Open the synth"): all its knobs, like on the machine;
- in the **SOUND screen**: the same settings in tabs, at the bottom of the screen, with the presets;
- as **pedals** ("FX · Effects" on the panel): the track's 4 insert effects as a pedalboard, over the instrument (Esc goes back to the instrument); the **EFFECTS screen** shows the same effects in signal order.

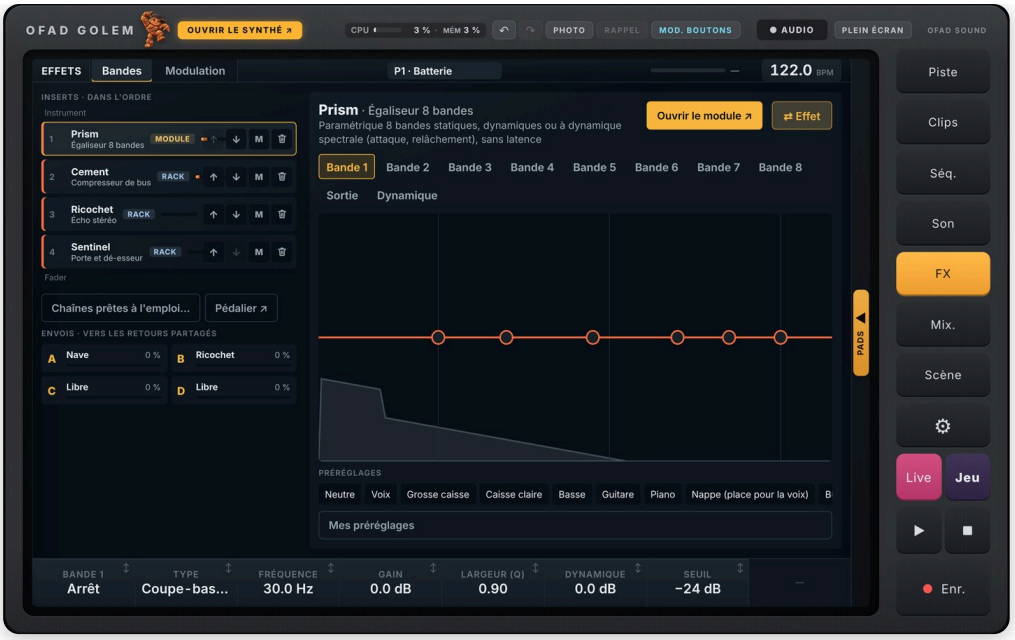
Everything is linked: turning the cutoff on the panel also turns it in the SOUND screen, an effect changed as a pedal also changes in the EFFECTS screen. Step locks, scenes, modulations, macros and MIDI learn work the same way from all three views.



SOUND screen: the instrument settings in tabs, with its presets



"FX · Effects" from the panel: the track's effects as pedals, over the instrument



EFFECTS screen: the channel's 4 insert effects, in signal order

The curves of the SOUND screen can be adjusted with a finger (1.1). Envelopes, filters and waveforms follow the real values, and **their points can be dragged**: attack, decay, sustain, release (and the rate and level envelopes of the FM synths), cutoff across and resonance up and down for a filter, pulse width of an oscillator. A dragged point writes its setting **where the encoders write** (base sound, step lock, scene). While playing, a **point runs along the envelope**; with a modulation (LFO…), a ghost point and a dotted curve show the sound being played, and a line shows the whole range. The "M1 +30 %" depth under an encoder is set by dragging it. **Settings › Display**: animated or still curves; above 90 % audio load for 1.5 s, the animation switches itself off to leave room for the sound (message, turn it back on by hand).



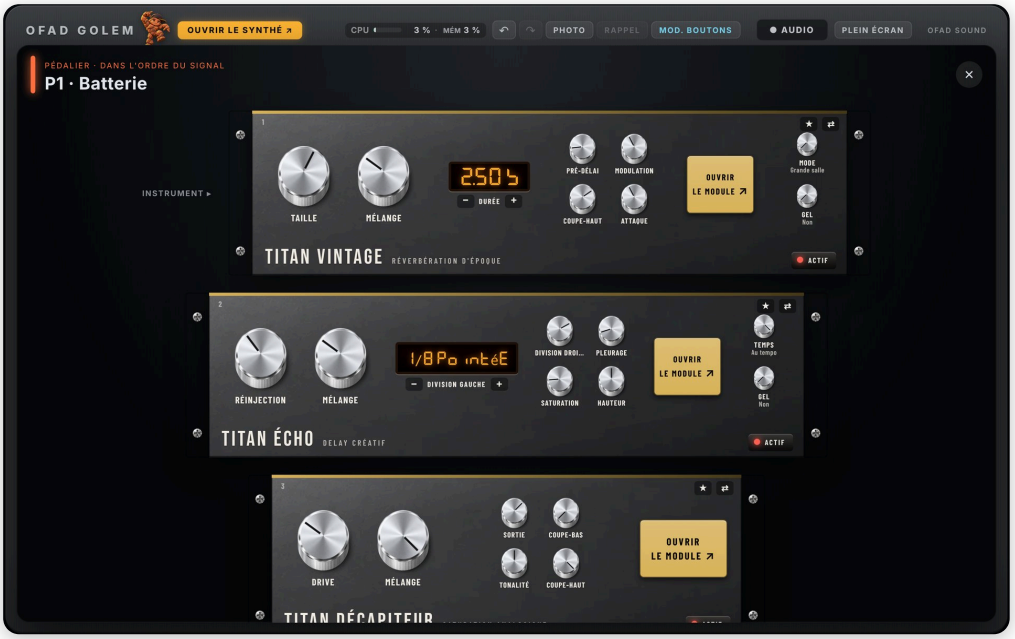
SOUND screen: drag the envelope points, the encoders follow

Golem Titan FX (1.1): nine high-end effects, a compact black-metal face in the pedalboard (with the time settings, sync and freeze) and a **full interface** through "Open the module" (full screen on a phone):

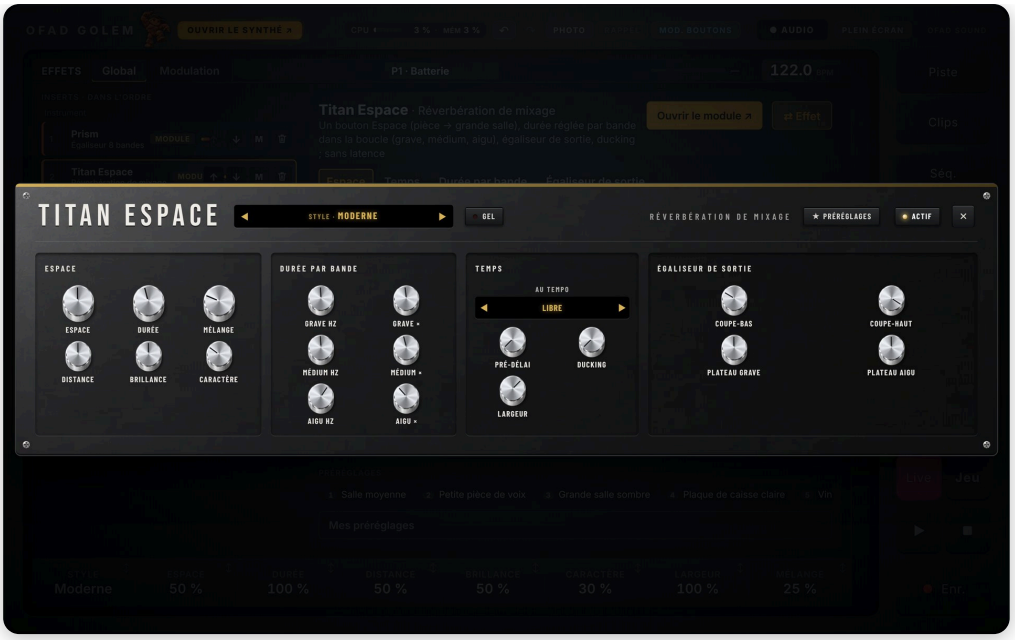
Effect	What for
Titan Vintage	Digital reverbs from the 70s–80s (12 models) and 3 vintage colours
Titan Espace	Mixing reverb: one "Space" knob (room → large hall), decay per band, output EQ, ducking, freeze
Titan Limiteur	Mastering limiter: 8 styles, 0–5 ms lookahead (compensated), true peak, oversampling, unity-gain listening
Titan Écho	Creative echo: two tape lines, ping-pong, "tape" or "stretch" time changes; filters, saturation, lo-fi, diffusion and pitch in the loop
Titan Décapiteur	Analogue saturation: 5 circuits (tube tape machine, console, germanium, triode, pentode), thump, high cut
Titan Saturne	Multiband saturation: 1 to 4 bands, 16 styles per band (tubes, tapes, transformer, amps, fuzz…)
Titan Phaseur	Phaser from 2 to 24 stages, 8 circuit styles; LFO, rhythm, steps, envelope
Titan Flanger-Doubleur	Two-tape flanger (through-zero) and voice doubler
Titan Filtre	2 filters (series, parallel, stereo), 8 types, 6 to 48 dB/oct, self-oscillation, 9 styles; LFO, rhythm, follower

Restoration (1.1): seven tools to clean up sound, to insert like any effect or, better, in the **sample editor** and the **AUDIO** screen (offline processing, maximum quality): **Denoiser**, **De-hum** (50 / 60 Hz mains), **De-click**, **Spectral de-esser**, **De-plosive**, **De-clip** (redraws clipped peaks) and **De-reverb** (removes room echo). Each tool has a **Listen** to what it removes.

Latency: some tools (denoiser, lookahead limiter…) delay the sound. GOLEM **compensates by itself**: the other tracks, the groups and the returns wait, nothing drifts. The **Latency xx ms** indicator in the top bar shows it; touch it to see which tools, on which track.



Golem Titan FX in the pedalboard: adjustable displays, sync and freeze on the face



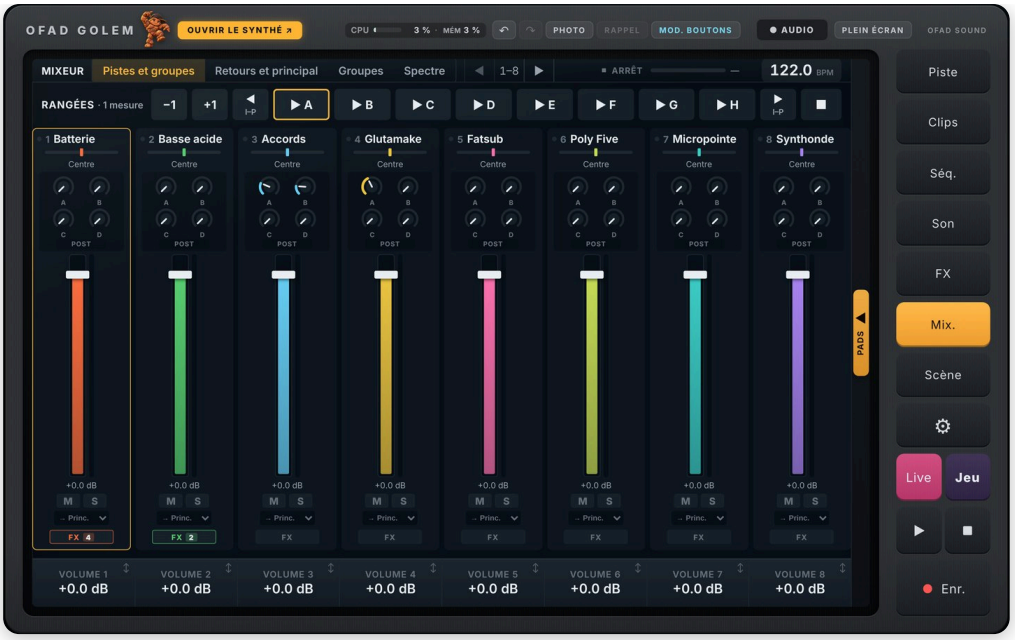
Titan Espace, full interface



Restoration: Denoiser and De-hum as pedals

6. Mixing

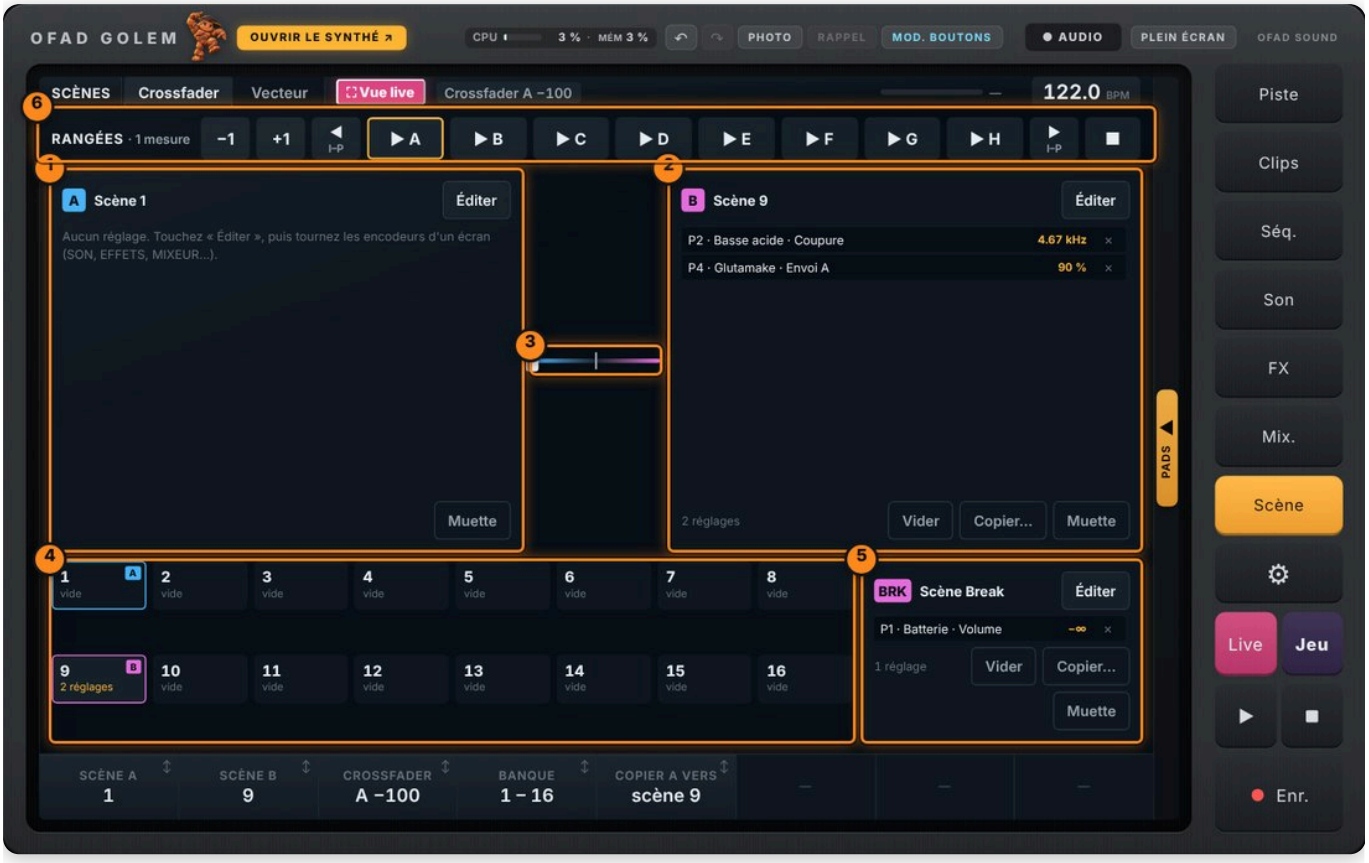
MIXER screen: volume, pan, mute (M), solo (S), sends to returns A to D, 7 groups and the master. The **sends A, B, C, D** are knobs (tracks: in place of the EQ; groups: EQ and sends), also in the live view; **long press** (right-click): exact value. A strip's EQ acts after its insert effects; on the **master** it acts before them: a limiter in the last insert stays the last stage of the sound and keeps its ceiling.



MIXER screen: tracks and groups, returns and master

7. Scenes

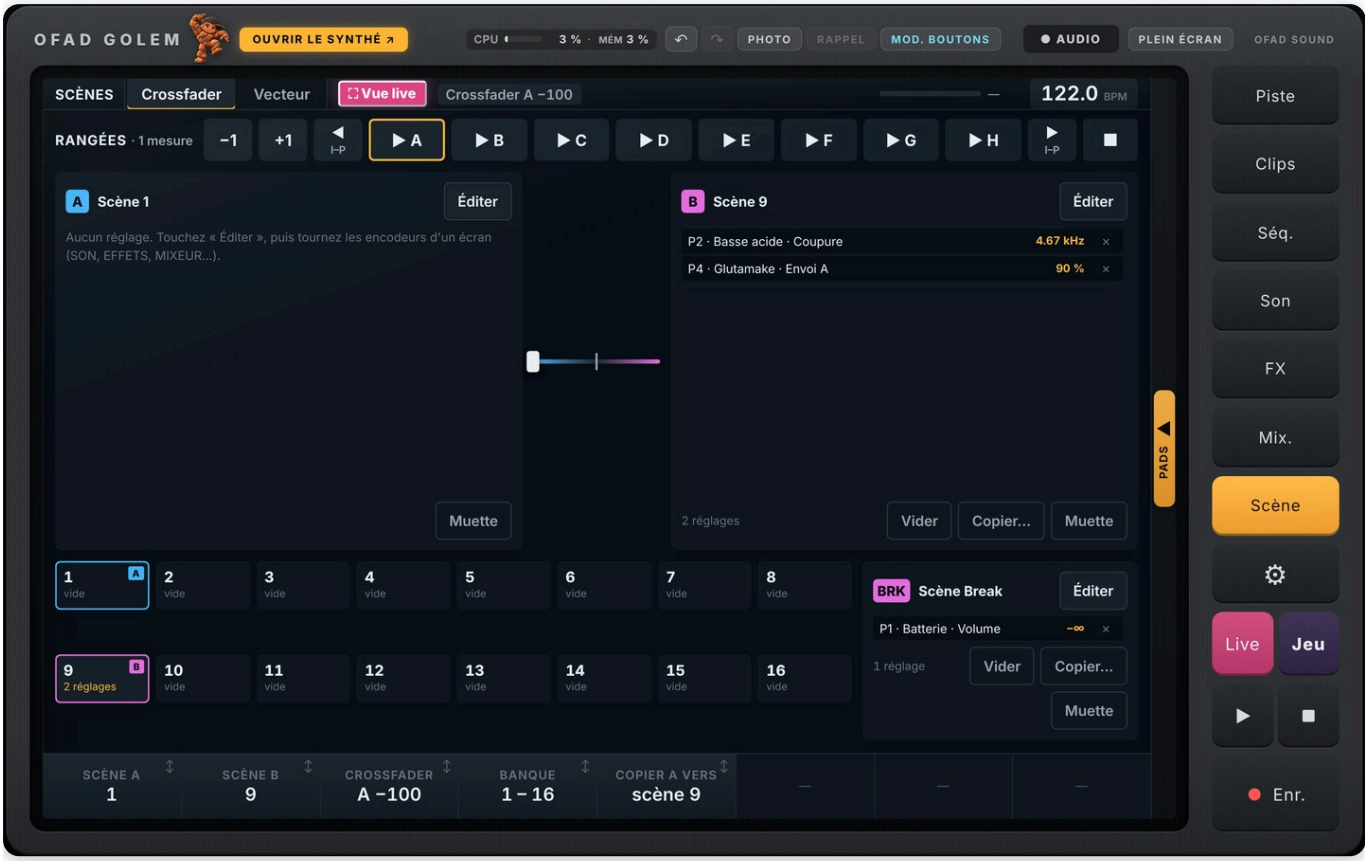
A **scene** is a list of settings (from any track, effect or mixer) with their values. There are **32 scenes** (two banks of 16) and a **Break scene**. Two scenes sit on either side of the **crossfader**: scene **A** on the left, scene **B** on the right. As you push the crossfader, the settings move **smoothly** from A's values to B's (stepped settings switch in the middle). Only the settings stored in A or B move: everything else stays as it is.



SCENES screen

1. **Scene A**: "Edit", then turn knobs on any screen (SOUND, EFFECTS, MIXER, panel, pedals): each setting you touch is added to the scene with its value. "Done" to exit. Values can also be adjusted in the list (drag), × removes a setting.
2. **Scene B**: the same, on the other side.
3. The A ↔ B **crossfader** (touch or drag; also in the Play view, the live view and from the MIDI controller).
4. The **scene grid**: touching a scene places it **on the side you hear** (where the crossfader is); with **Shift**, it goes **to the other side**, ready to be brought in with the crossfader. Right-click: mute scene. The "Scene A" / "Scene B" values at the bottom also choose the scene on each side.
5. The **Break scene**: applied while **Break** (or the **B** key) is held.
6. **Rows A to P**: launching a row also launches its **linked scene** (set in CLIPS).

Playing several scenes in a chain: crossfader on the left on scene 1. **Shift** + scene 2: it goes to B. Push the crossfader to the right: the sound glides from scene 1 to scene 2. **Shift** + scene 3: it goes to A (the side you no longer hear). Bring the crossfader back to the left: the sound glides to scene 3. And so on, as many scenes as you like, without a gap. Crossfader movements are recorded in the song ("Rec motion"). From a MIDI controller, Program Change 0 to 31 launch scenes 1 to 32 (on a channel no track is set to).



SCENES › Crossfader (PC): scene 1 in A, scene 9 in B, the crossfader between them

The vector pad (SCENES screen, **Vector** tab; the **Crossfader** tab keeps the A / B crossfader). Four scenes, one per corner (**A B C D**, chosen in "Corners"), blend according to the position of the dot on the pad: in the centre, all four in equal parts; in a corner, that corner's scene.

- **Pad active**: the pad replaces the A / B crossfader. Touching the pad activates it and places the dot; double-click: back to the centre. **X** and **Y** are also set at the bottom of the screen.
- **Motion**: the dot moves by itself around its place — Circle, Infinity, Diamond, Square, Triangle, Bounce, Random, or **Drawn** ("Draw a shape": draw it with your finger). The chosen shape is shown on the pad.

- **One cycle in**: from 1/2 to 128 bars, locked to the tempo. **Size**: the extent of the motion. **Height**: flattens or stretches the shape (from 5 % to 200 %): the circle becomes an oval, the square a rectangle, the infinity sign squashes or stretches. **Rotation**: turns the shape (0°, 45°, 90°, 135° in one touch; Infinity turns vertical at 90°). A drawn shape keeps its place and direction.
- **Shaping the motion on the pad**: dragging **moves** it (without jumping). On PC, **right-click drag** up or down: size; left or right: rotation; **wheel**: size, **Shift + wheel**: rotation. With fingers, **two fingers**: pinch for size, twist for rotation.
- All these values can be handed to an assignable knob or to the MIDI controller.



SCENES › Vector (PC): the "Infinity" motion travels through the four scenes

Live (raspberry button on the rail, or ⛶ Live view next to the Crossfader / Vector tabs): the whole performance on a single screen, the showcase for playing live. At the top: Back, ● ▶ ■, **Crossfader** or **Vector** mode (with, in Vector, the motion, cycle length, Draw), the selected track and **Open the instrument**; then the **16 assignable knobs with the name of their function**; the **rows** (see below).

- **Crossfader mode**: scene A and its settings, a big crossfader (A, B, double-click: middle), scene B; next to it, the scene grid (banks 1–16 / 17–32, touch: launch) and the Break scene.
- **Vector mode**: the square pad and all its settings (corners, motion, length, size, height, rotation, drawing). **Crossfader** and **Vector** choose the view: switching to Crossfader stops the pad, switching to Vector does not start it. You set it up (you can even place the dot with your finger)

without the sound changing, then **Pad stopped / Pad active** (green while it plays) starts or stops it.

- The live view's **scene cards** are for playing: the scene name and its settings. You edit them beforehand, in the SCENES screen.
- **Console** (PC): the 16 tracks with volume fader, peak meter, **FX** (the track's effects, as pedals), **M / S**; **Tracks / Groups** shows the mixer groups, and in Tracks view the groups in use take the place of tracks with no instrument. Touching a track's name or fader selects it (Open the instrument, effects, knobs to assign); ✕ in the instrument or the effects brings you back to the live view.
- At the bottom: the **position** (bar, beat, loop, song row), **Fill, Break, Break 1 bar, Shift** and **Photo / Recall / Knob mods**.
- On a phone, the console becomes the track list (name: select, M / S).



Live view · Crossfader



Live view · Vector



Live view on a phone

8. Song

CLIPS › Song screen: **rows** played one after another in song mode. "Rec performance" turns your playing into rows; **REC** (or "Rec motion") records the movements: knobs, faders, EQs, mutes, solos, groups, crossfader. In song mode they go to the song; in loop mode, a track's settings go into its pattern's loop.



CLIPS › Song: the rows (pattern of each track) and, below, the recorded curves

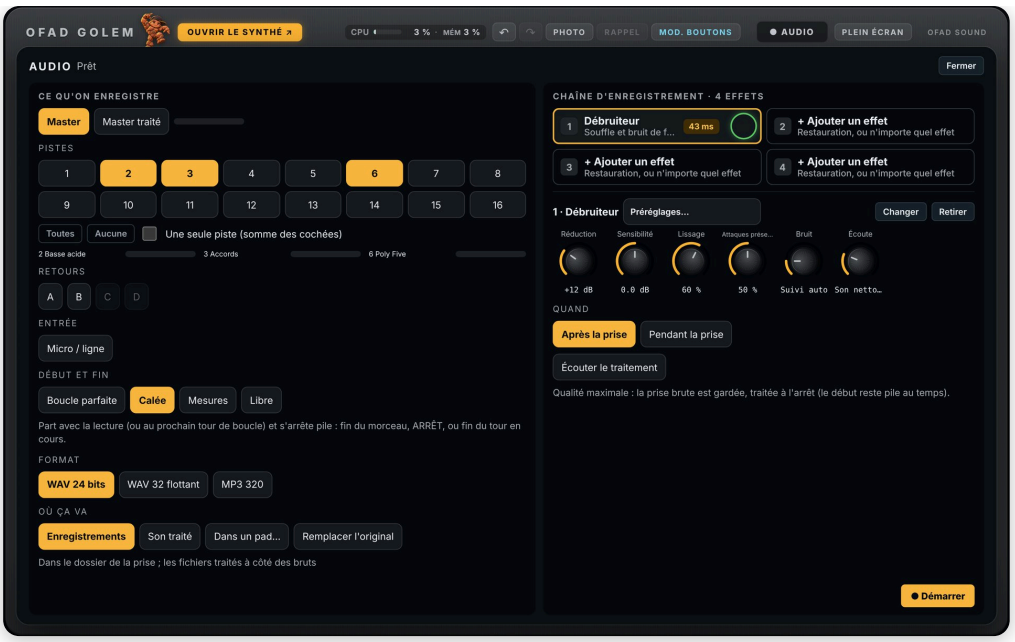
9. Audio arrangement

CLIPS › Audio screen: place the playhead, "● Record a take", play (the take starts at the beginning of the next bar and lands at the playhead); at the end, each track becomes audio clips, to split, move, duplicate; **Arrange mode** to hear them; WAV or MP3 export of the finished track.

Recording a loop, right on the bar: place the loop on the ruler (right-click + drag, "Bar" snap), turn **Loop** on, then "● Record a take". The take starts at the beginning of the next bar, lasts **exactly** the length of the loop, **stops by itself** at its end (to the sample) and lands at the start of the loop: each track played becomes a clip exactly the length of the loop. No more stopping by hand at the right moment.

AUDIO screen ("● Audio" button at the top, 1.1): a whole screen to record and process. A **long press** on "● Audio" starts right away with the last settings.

- **What is recorded**: the **Master**, the **Processed master**, ticked **tracks** (one file per track, or "A single track" = their sum), returns A to D, the **input** (mic / line, with its meter). While listening, you only hear what you record.
- **Start and end**:
- **Synced**: the take starts with playback (or at the next loop cycle) and stops **exactly**, to the sample: end of the song, STOP, or end of the current cycle when you press "■ Audio" again;
- **Perfect loop**: one run-up cycle plays first (echoes and compression settle), then the take lasts **exactly one cycle**, with a crossfaded join: the loop replays **without a click** (prefer WAV);
- **Bars**: exactly N bars; **Free**: everything, even without playback, until "■ Audio".
- **Format**: 24-bit WAV, 32-bit float WAV, 320 kbit/s MP3 (no silence added at the start).
- **Recording chain**: **4 effects** of your choice (Restoration or any effect), applied **after the take** (maximum quality; the raw take is kept) or **during the take** (file processed and realigned as it is written). **Listen to the processing** sends the output through the chain, latency shown.
- **Where it goes**: **Recordings**, **Processed sound** (files named "Track – 128 BPM – 4 bars – processed"), **Into a pad** (first free slot of the chosen track), or **Replace the original** (can be undone).



AUDIO screen: sources, start and end, format, destination, 4-effect chain

Takes are stored in **Library › My recordings**.

Exporting the arrangement: "Export" offers **the loop** (the ruler area) or **the whole arrangement**, exactly to length, as 24-bit WAV or 320 kbit/s MP3. **Several clips at once**: tap a lane's name to select all its clips, **Ctrl+A** selects everything, and dragging the edge of a selected clip adjusts every selected clip.

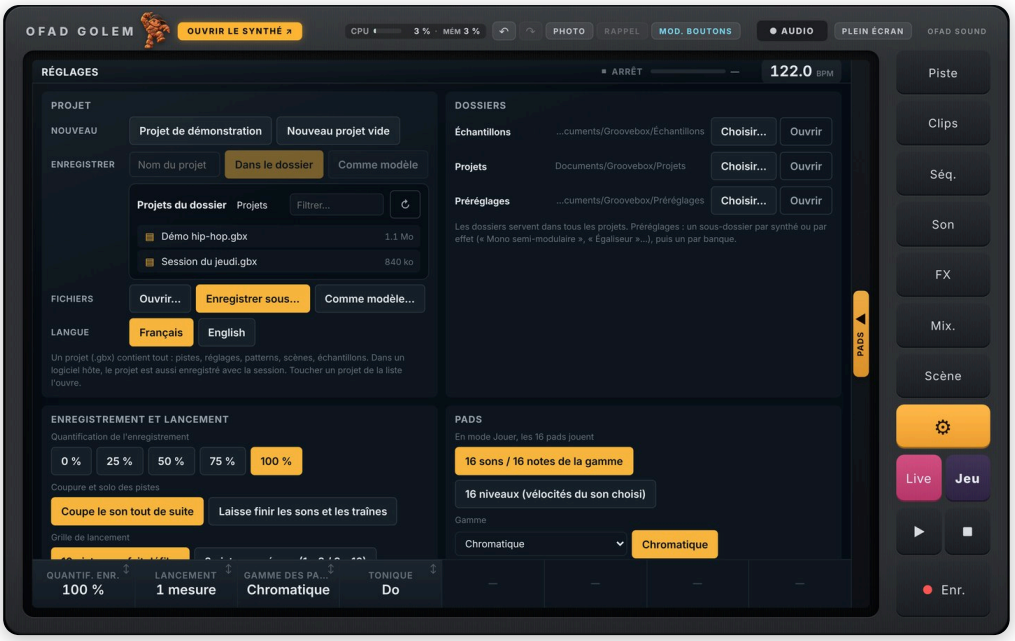


CLIPS › Audio: the clips of each track, the ruler, the playhead; below, the selected clip (volume, fades, split, copy, duplicate)

10. Saving

SETTINGS › Project screen: Save (a .gbx file with sounds, samples, sequences, scenes) or "As template" (settings and sounds, without the writing). **Techno demo**, **Tribe demo**, **New empty project** and **Open...** light up yellow when touched, and a message confirms the loading. The Tribe demo (1.1, 180 BPM) has 5 instruments, 4 scenes, the vector pad on a circle and a short song (intro, build-up, break, drop): just press Play. The Techno demo is deliberately light (7 instruments, 11 effects) to run on every phone. It also shows the vector pad in motion (Circle, one cycle in 16 bars) and assignable knob

1 linked to the filter cutoff of Kick Killer. The audio arrangement is stored in the "<project name> - audio" folder next to the file.



SETTINGS screen: project (demo, empty, open, save, template), folders, recording, MIDI

On tablets and phones: projects are saved and opened **inside the app** ("Saved projects" list, tap to open; any name, otherwise the date). **Export a copy...** and **Import a file...** use the system file picker (Downloads, Drive...) to exchange a project.

Undo and photo: ↶ ↷ (Ctrl+Z, Ctrl+Y) undo up to 50 steps. **Photo** freezes all settings; **Recall** goes back to it in one gesture (and can be undone with Ctrl+Z).

11. Samples and Library

Do this first: tell GOLEM where your sounds are. The Library (pads, samplers, "Sample...") opens in the samples folder; set it right and you land straight on your sounds.

- **On a computer (Windows, Mac)**: **SETTINGS** screen (⚙ on the rail) › **Folders** block › **Samples** row › **Choose...**, then pick the folder where you keep your sounds (for example `Documents\My samples`). It is kept for every project; **Default** goes back to `Documents\Groovebox`, **Open** shows it in the file explorer. Do the same for **Projects** (Open... and Save... start there) and **Presets** (one subfolder per synth or effect, then your banks).
- **On a phone or tablet**: the system does not let the app read a folder freely, so the sounds are **copied** once. **SETTINGS › Folders › Samples › Personal folder...**: pick your sound folder, everything in it (subfolders included) is copied to **Library › Custom**; running "Personal folder..." again only adds the new ones. **Import files...** copies a few chosen sounds to **Library › Imported**.
- Formats accepted everywhere (pads, samplers, import, Library): **WAV, AIFF, FLAC, OGG and MP3**.
- **"Sample..."** (a sound taken from outside the Library): the sound is loaded at once on the pad or sampler, and **saved inside the project** (.gbx) at the next save — the project stays complete even if the original file moves.
- **On a computer**, the original file stays where it is: nothing is copied. To find it in the Library, put it in your samples folder (see above).
- **On a phone or tablet**, it is also copied to **Library › Imported**, so you can pick it again later without the file picker.

- **Settings › Personal folder…**: copies every sound of a folder of the phone or computer (subfolders included) into **Library › Custom**; doing it again only adds the new ones.
- **Library › My recordings**: your audio takes ("● Audio" button).

Sample editor (1.1): the same everywhere — sampler panels, drum kit pads, TRACKS screen (**long press** or **right-click** on a track that plays samples), a take sent into a pad. Waveform and selection; **Invert**, **Normalize**, **Fade in / out**, **Trim**, **Silence**; and **Process…**: a chain of **4 effects** (Restoration included) computed **offline** (no latency while playing), **previewed**, then applied **in place of** the sample — the pads and patterns that play it don't change. **Undo** keeps 8 versions.

12. Three special instruments

- **Glutamake** (wavetables and physics): two wavetable oscillators (A/B mix, morphs, modifiers); **Physics** page: a **ball** rolls on a surface (tilt, friction, bump, edges that absorb or bounce back). **Hold it with a finger** and **release it with a flick**, or let each note throw it; its motion (X, Y, distance, angle, hits) feeds the matrix. **Motion sequence** page: 4 lanes of 16 steps (loop, direction, division, smoothing, probability).
- **Good Doctor** (hybrid wavetables): three oscillators **A, B, C**; for each, the **Table**, the **Position**, **two warps in a row** (Warp 1 and 2 with their amount), the start **Phase** and its randomness, and the **Output** (both filters, F1 only, F2 only, or **Direct** for a clean low end). Sub oscillator and noise with their own output; **two filters** (14 types) in **series** or **parallel**; per-voice distortion; 32-slot matrix and 8 macros.
- **Pieuvre 8** (performance sampler with 8 lines T1 to T8): the last note played (keyboard, pads, MIDI while stopped) is written to the steps you place; **lengthen a note** by dragging a step's right edge; a selected step opens its **menu** (note, velocity, length, offset, condition, probability, repeats, locks).



Good Doctor: three wavetable oscillators, two warps each, two filters



Glutamake: the ball, the wavetables, the filter

13. FM Player: M-VAVE FM-1 firmwares

The **FM Player** runs inside GOLEM the **firmwares** (`.fwsc` files) written for the M-VAVE **FM-1** synth box: its processor is **emulated**, and each firmware becomes a GOLEM instrument, played by the sequencer, the piano, the pads and MIDI, with its screen and controls on the panel. It also works inside music software (VST3 / AU plugin).

- **15 free firmwares included**, copied once to **Library › Samples › Firmwares** at first launch: B1, FIZZ, FM-106, FMini-1, Felucca, Frodhes, GR!MM, JIANT FM, MYCELIUM, Melodee, OberVave, SLOOP, X0X, sloopDX, zp12 (authors, GPL-3.0 / MPL-2.0 licences and sources in `FIRMWARES-LICENCES.md`). **Firmware…** loads another `.fwsc` file (your own, or a community firmware).
- **The machine**: the live screen, KNOB 1 to 4, SELECT, PRESETS, ALGORITHM, the MASTER pot, the 14 function keys (FX… REC, OCT − / +) and the 27 small keys with their LEDs (in the firmwares' SEQ mode they act as steps). Knobs are set by dragging up or down, or **by turning** around the knob.
- **Tracks T1 to T4** (under the piano, as on Pieuvre 8): each T plays on **its own MIDI channel** — T1's channel ("T1 channel", Auto by default), then the next ones. Multi-track firmwares (SLOOP, sloopDX, JIANT…) thus play their 4 tracks. The selected T is the one you write (**Acid** and **Locks** rows, **SEQUENCE** screen with 4 lines and "T in notes ♪") and the one played by the piano, the pads and MIDI.
- ⤢ **Big**: the machine on the whole screen area (phone and tablet), with GOLEM's piano and T1 to T4 at the bottom, **M / S** above each T, the chosen T's **step row** (locks, bars, shifting the T or everything; Copy / Paste / Clear and **Locks** under the screen) and **Library** to switch firmware. ✕ **Close** goes back to the panel.
- **Background of your choice**: the dots next to "Big" — **Amber**, **Night**, **Emerald** or **Plain** (kept on the device).
- **Settings**: Level, Transpose, **Lead** (how far ahead the emulation runs, 40 ms by default: raise it if the sound crackles on a slow device), **MIDI clock** (the firmwares' sequencers follow GOLEM's tempo), **Notes played by** (Auto, MIDI, or the box's keys for firmwares that don't read MIDI).
- **The box's 8 controls on your controller**: KNOB 1 to 4, SELECT, PRESETS, ALGORITHM and MASTER are also instrument settings. **Long press** (or right-click) a knob: **MIDI learn**, assignable

button, modulation, as on every synth. On the **SOUND** screen, the **Box** section puts them under the **8 encoders**. The knobs go from **0 to 100 %**: by finger, the whole travel in one drag (or by turning).

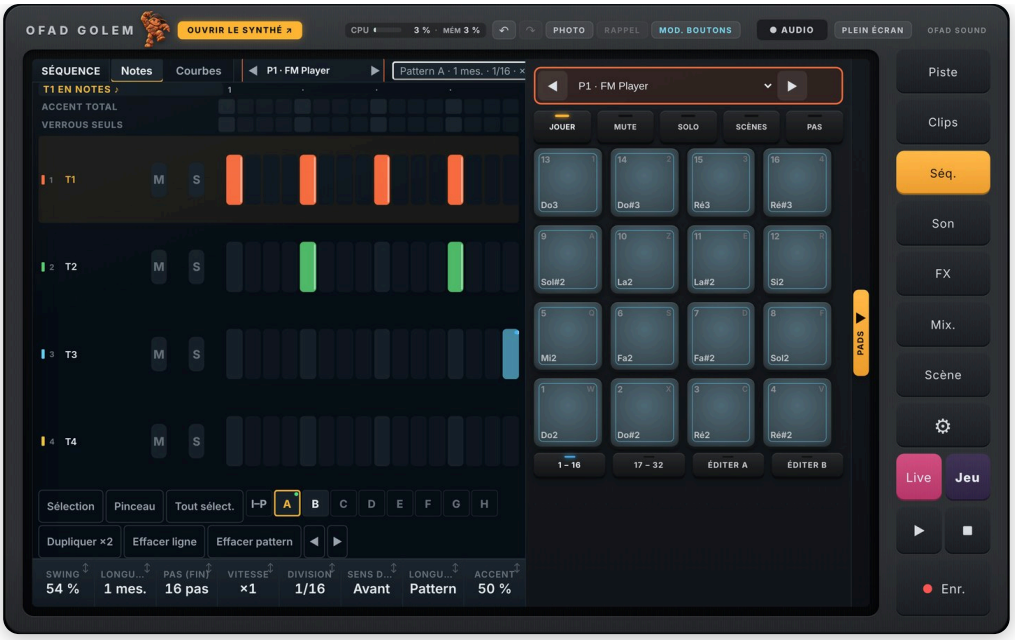
- **Knob travel** (Firmware panel, folded): how many firmware clicks go from minimum to maximum, knob by knob — **128** for KNOB 1 to 4, **4** for ALGORITHM (a quarter turn per algorithm), 256 for SELECT and PRESETS. Firmwares don't all count the same: if a setting reaches its end too early, lower its travel. At **0 %** and **100 %**, GOLEM re-aligns the firmware on its end stops.
- **The box's 14 keys** (FX, SCL, ENV, LFO, EDIT, GLO, HOME, SAVE, ARP, SEQ, PLAY, REC, OCT − and OCT +): **right-click** or **long press** to learn them — a controller button then presses the box key (it lights up blue while learning).
- **MIDI controller controls** that GOLEM has not learned go straight to the firmware.
- **Paused** after 20 s without playing (saves battery): a note or a key wakes it up.



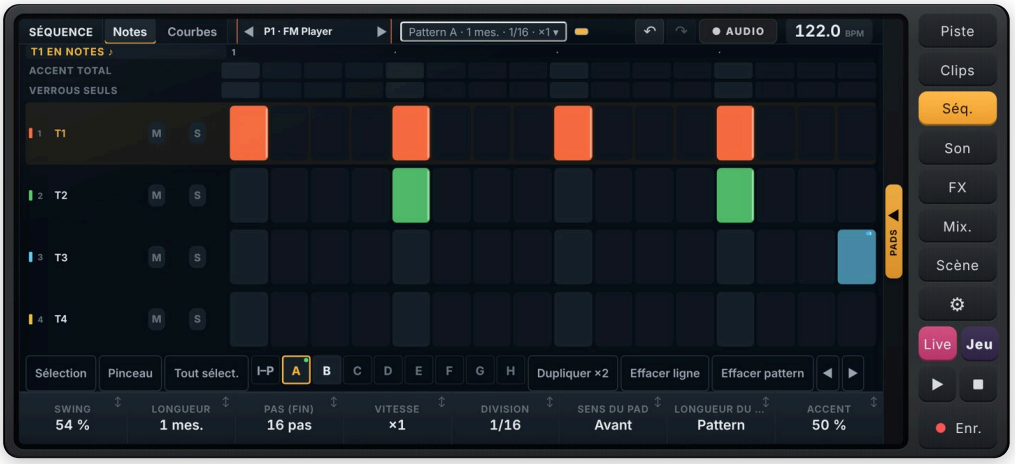
FM Player (Amber background): the machine, T1 to T4 and the Acid row under the piano



Big mode on a phone (Night background): screen, knobs, piano and T1 to T4



Sequence screen: one line per T track (one firmware MIDI channel each), pads folded out to play



The same sequence on a phone

M-VAVE and FM-1 are trademarks of their respective owners. OFAD GOLEM is not affiliated with, endorsed or sponsored by M-VAVE; these names are used only to state compatibility with firmware written for the FM-1. Emulation is based on fm1-emu (Joakim Eriksson, MIT licence). No original M-VAVE software is included: the bundled firmwares are free community works, with their authors and sources credited.

14. MIDI track: playing an external synth or drum machine (1.1)

The **MIDI track** instrument (**External MIDI** family, TRACKS screen) makes no sound: it **plays the connected device** (synth, drum machine, module). Everything a track has works on it: sequencer (Acid, Locks, SEQUENCE), locks, scenes, modulations, launching, song, piano, pads and controller.

- **The top bar** (lists): **Device** (the MIDI output; ↻ reads the list again; the link type — USB, Bluetooth, network — is shown when the system reports it), **Send** (channel 1 to 16), **Receive** (the channel that plays this track from a keyboard), **Bank** and **Sound** (CC 0, CC 32 = 0, then program change, like the remote). On a phone, **Bluetooth…** opens pairing.
- **32 knobs in 4 sections of 8**: each knob sends its value as a **CC**. **Touch a section title to rename it** ("Filter", "Envelopes"…): the names are kept in the project. A knob's small **⚙ wheel** opens its settings: **Learn** (touch it, turn the synth's knob: its CC number and channel are set), **CC** (0 to

119), **Channel**. Nothing is sent on loading: the device's sound is not overwritten; "Send at start" sends the values again at playback.

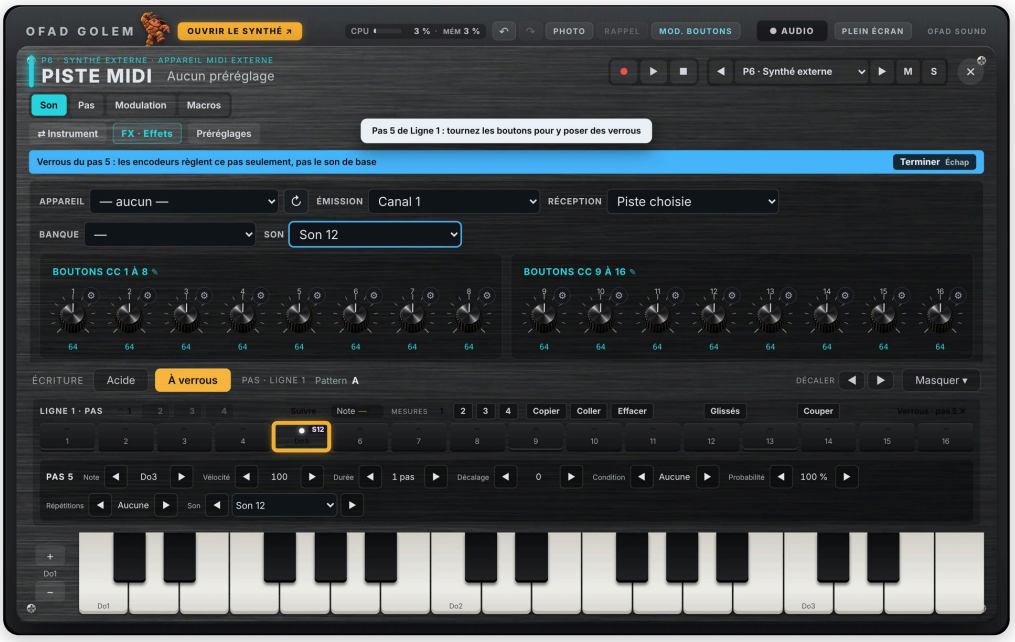
- **CC numbers and channels are fixed**: set them once and for all. A scene, a lock, an LFO or a curve can **never** change them (only the knob's value moves), and CC 120 to 127 (device commands: all off, reset) are never sent.
- **16 pads**, set up with their ⚙ wheel like those of the remote: **CC held** (127 while pressed), **CC on / off**, **CC steps** (each press moves to the next value, 2 to 8 values) or **Note**.
- **Sequencer**: the step row is under the keyboard as on the synths. Long press on a step, then: turning one of the 32 knobs sets a **CC lock**; touching a pad **locks** it on that step (pressed for the step's note length, every time round); the **Bank** and **Sound** lists set the **step sound** (the program change goes out just before the note).
- **MIDI effects**: arpeggiator (20 modes, octaves, rate, length, hold, swing), chord and spread, scale and root, note echo (fading repeats, transposition), probability, fixed or random velocity, transposition.
- **Mute and solo**: notes in progress stop, nothing goes out while the track is muted.
- **Output**: in the application (PC, Mac, phone), the output chosen in "Device" (one output shared by all MIDI tracks, each on its own channel). In music software, the **plug-in's MIDI output**.
- **Possible links**: **USB** everywhere; **Bluetooth MIDI** on phones and tablets (Bluetooth...) and on Mac (the system pairing window); on Windows PC, through a bridge driver that creates a standard MIDI port; **network** through a driver such as rtpMIDI, which shows up as a device in the list. What the device sends back to GOLEM does not replay the track (no MIDI loop).



MIDI track: the top bar, the 32 knobs in 4 renamed sections, the 16 pads



A knob's ⚙ wheel: Learn, CC number, channel



Step sound: step 5 sends sound 12 to the device just before its note

MIDI controller

Plugging in a controller: plug and play

- **USB**: plug in the keyboard or controller, it is recognised by itself, with nothing to set up — in the PC application as on an Android phone or tablet (with a USB-C "OTG" cable or hub). You can plug it in or unplug it while the Groovebox is running.

- **Bluetooth (Android)**: Settings › "**Pair a Bluetooth controller...**". The first time, Android asks for Bluetooth permission; then choose your wireless MIDI keyboard or controller from the list: it plays like a USB controller.
- **On Mac**, pair the device in the system's Bluetooth MIDI window (Audio MIDI Setup).
- **On PC**, a Bluetooth controller only works if Windows presents it as a MIDI port (bridge driver): USB is the simplest.
- **In a host application** (VST3): the host receives the controller and sends it to the Groovebox.
- Notes play the selected track (or the tracks set to a channel, Settings › MIDI); on a drum kit, notes 36 to 51 play the 16 pads. On **Pieuvre 8** and the **FM Player**, the keyboard plays the chosen T **across all its octaves**, and each T can have **its own channel** (TRACKS screen, "T1 ... T8"): an external keyboard or sequencer then plays several Ts at once. The mod wheel, pitch bend and sustain pedal follow the track being played.
- Phone with a single USB-C port: a USB-C hub with charging (PD) lets you plug in the keyboard, the USB audio output to the mixing desk and the charger at the same time.

Assigning the controller's controls

- Right-click (long press) on a setting › **Learn MIDI**, then touch the control on the controller. This also works for the **values at the bottom of the screens** and the global settings: tempo, swing, crossfader, scenes A and B, vector pad (X, Y, motion, size...) and macros.
- The **16 assignable knobs** can be learned too: map a row of knobs on the controller to knobs 1 to 16, then hand them whatever you want (Assign to...).
- Right-click on the buttons (M / S, rows, Photo, Recall, transport...): assign them to the controller.
- Settings › MIDI: notes channel, commands channel, a channel for each track.
- **Changing sounds over MIDI**: on a track set to a channel, bank select (CC 0 / CC 32) + program change call the sound number shown in front of each preset ("Init" = 1, your sounds after them); CC 102 to 105 change the sound of the effect in inserts 1 to 4. Details are in the **MIDI implementation** document (MIDI-IMPLEMENTATION).
- In the Android app: Settings › "**Pair a Bluetooth controller...**" (wireless MIDI keyboard or controller); a USB keyboard is recognised by itself.

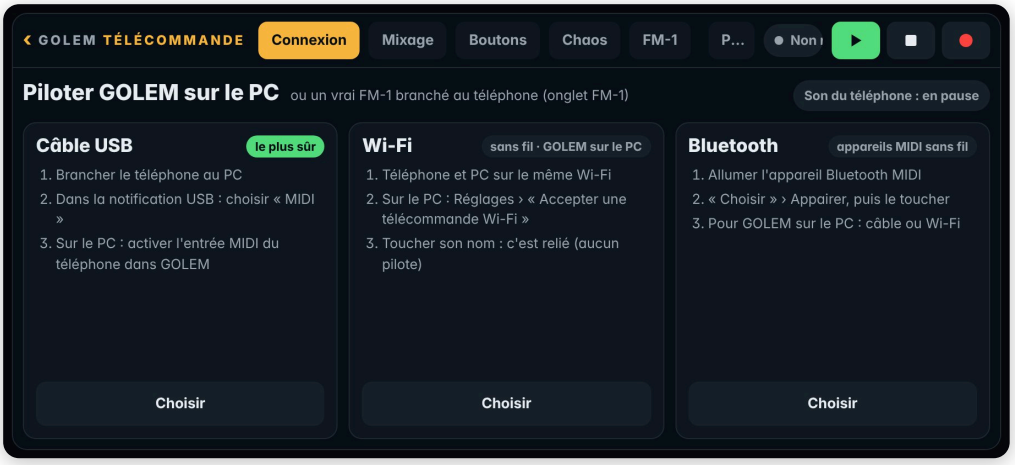


Long press (right-click) on a bottom value: exact value, modulation, Learn MIDI, knob 1 to 16

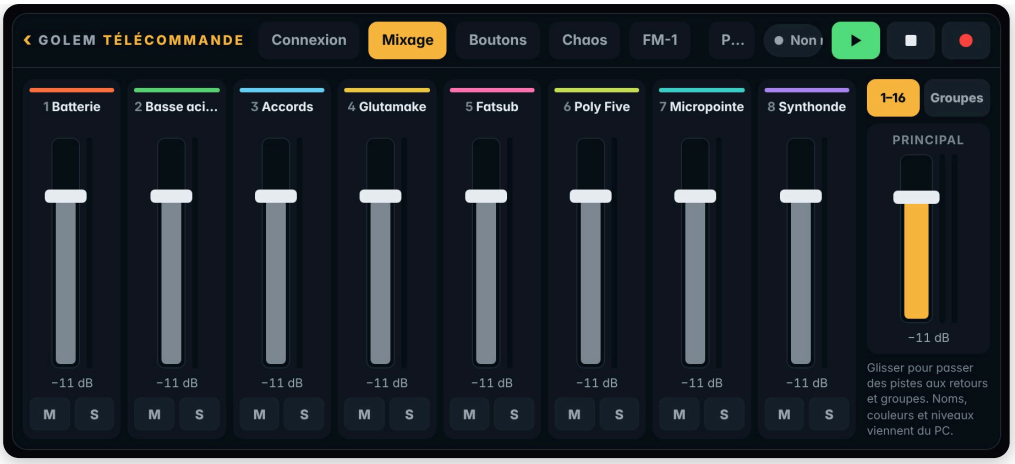
Remote: the phone controls GOLEM (1.1)

A phone (or a tablet) running GOLEM becomes the **remote** of GOLEM on the PC, of an FM-1, or of another MIDI device. Way in: the **Remote** button on the home screen, or Settings › MIDI controller › "Remote mode..."; **‹ GOLEM** goes back to GOLEM without restarting. The phone's sound is paused by default.

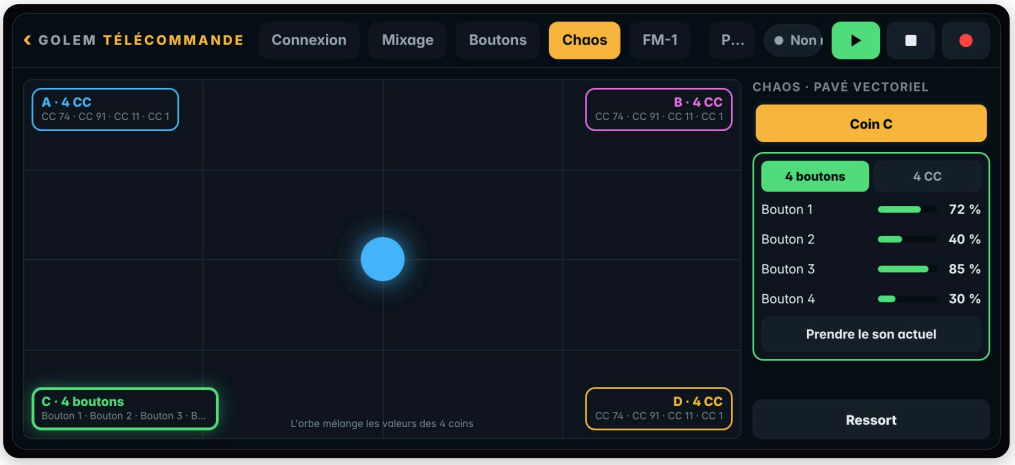
- **Connection**:
- **USB cable** (the safest): connect the phone to the PC, choose "MIDI" in the USB notification, then turn on the phone's MIDI input in GOLEM on the PC;
- **Wi-Fi** (no driver): phone and PC on the same network; on the PC, Settings › "**Accept a Wi-Fi remote**"; on the phone, touch the PC's name. Local network only, one remote at a time;
- **Bluetooth**: for a wireless MIDI device (Pair, then touch it).
- **Mix**: the 16 tracks (2 pages of 8), volume, M, S, groups, master; names, colours and levels come from the PC.
- **Knobs**: GOLEM's 16 assignable knobs (name and value sent back by the PC), crossfader, Fill, Break, Shift, play, stop, record.
- **Chaos**: the vector pad, with the motion shape chosen on the PC. Each corner drives 4 GOLEM knobs or 4 CCs; "Take the current sound", **Spring** (the orb returns to the centre).
- **FM-1** (or any MIDI device): keyboard and wheels, pads (CC held, CC on / off, note), 8 CC knobs, sound − / +, bank, channel, firmware profiles, sending a .syx file. For GOLEM on the PC, the sound changes the instrument or the effect of insert 1 to 4 of the track.
- **Learn**: with a device connected, the small "Learn" button next to a CC setting; touch it, turn the device's knob (or play the note), the number and channel are set.
- **Profiles**: save the setup (knobs, pads, Chaos pad, device tab) under a name ("Bass synth profile"...) and recall it with one touch.



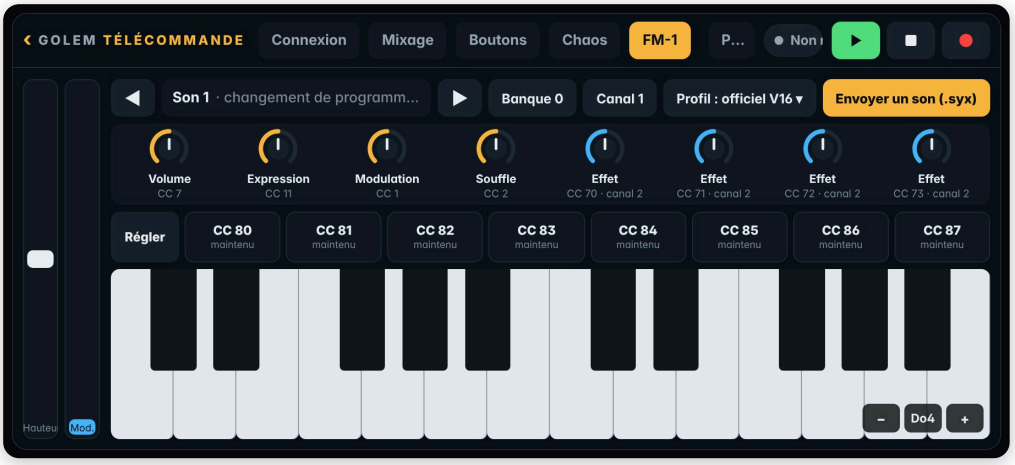
Remote · Connection



Remote · Mix



Remote · Chaos



Remote · FM-1

Where your files are

- Projects, samples, presets: Documents\Groovebox (folders can be changed in Settings).
- Audio recordings and arrangement exports: Documents\Groovebox\Enregistrements.
- Takes from a project's arrangement: the "<project name> - audio" folder next to the .gbx.